

UNIT 1 TOPICS

TOPIC 1 MOLECULES, TRANSPORT AND HEALTH

1A CHEMISTRY FOR BIOLOGISTS

- 1 THE CHEMISTRY OF LIFE 4
- 2 CARBOHYDRATES 1: MONOSACCHARIDES
AND DISACCHARIDES 8
- 3 CARBOHYDRATES 2: POLYSACCHARIDES 11
- 4 LIPIDS 14
- 5 PROTEINS 16
- THINKING BIGGER: TREHALOSE 22
- EXAM PRACTICE 24

1B MAMMALIAN TRANSPORT SYSTEMS

- 1 THE PRINCIPLES OF CIRCULATION 26
- 2 THE ROLES OF THE BLOOD 28
- 3 CIRCULATION IN THE BLOOD VESSELS 31
- 4 THE MAMMALIAN HEART 37
- 5 ATHEROSCLEROSIS 41
- EXAM PRACTICE 44

1C CARDIOVASCULAR HEALTH AND RISK

- 1 RISK, CORRELATION AND CAUSE 50
- 2 INVESTIGATING THE CAUSES OF CVDs 52
- 3 RISK FACTORS FOR CARDIOVASCULAR
DISEASE 54
- 4 DIET AND CARDIOVASCULAR HEALTH 56
- 5 DIETARY ANTIOXIDANTS AND
CARDIOVASCULAR DISEASE 59
- 6 USING THE EVIDENCE 63
- 7 THE BENEFITS AND RISKS OF
TREATMENT 65
- THINKING BIGGER: HEART FAILURE 67
- EXAM PRACTICE 70

TOPIC 2 MEMBRANES, PROTEINS, DNA AND GENE EXPRESSION

2A MEMBRANES AND TRANSPORT

- 1 CELL MEMBRANES 74
- 2 CELL TRANSPORT AND DIFFUSION 76
- 3 OSMOSIS: A SPECIAL CASE OF
DIFFUSION 80
- 4 ACTIVE TRANSPORT 83
- 5 THE NEED FOR GAS EXCHANGE
SURFACES 86
- 6 THE MAMMALIAN GAS EXCHANGE
SYSTEM 88
- THINKING BIGGER: ASTHMA AND
AIR POLLUTION 90
- EXAM PRACTICE 96

2B PROTEINS AND DNA

- 1 ENZYMES 100
- 2 HOW ENZYMES WORK 102
- 3 THE STRUCTURE OF DNA AND RNA 104
- 4 HOW DNA WORKS 108
- 5 THE GENETIC CODE 111
- 6 DNA AND PROTEIN SYNTHESIS 114
- THINKING BIGGER: RAW ENZYMES 117
- EXAM PRACTICE 120

2C GENE EXPRESSION AND GENETICS

- 1 GENE MUTATION 124
- 2 PATTERNS OF INHERITANCE 126
- 3 SEX LINKAGE 129
- 4 CYSTIC FIBROSIS: A GENETIC DISEASE 134
- 5 GENETIC SCREENING 138
- EXAM PRACTICE 142

UNIT 1 PRACTICALS -7

STUDENT BOOK TOPIC

TOPIC 1
MOLECULES, TRANSPORT AND HEALTH

TOPIC 2
MEMBRANES, PROTEINS, DNA AND
GENE EXPRESSION

IAS CORE PRACTICALS

- | | |
|-----|---|
| CP1 | Use a semi-quantitative method with Benedict's reagent to estimate the concentrations of reducing sugars and with iodine solution to estimate the concentrations of starch, using colour standards. |
| CP2 | Investigate the vitamin C content of food and drink. |
| CP3 | Investigate membrane properties including the effect of alcohol and temperature on membrane permeability. |
| CP4 | Investigate the effect of temperature, pH, enzyme concentration and substrate concentration on the initial rate of enzyme-catalysed reactions. |

Core Practicals →

Recommended Practicals →

UNIT 1 (TOPICS 1 AND 2) MOLECULES, DIET, TRANSPORT AND HEALTH

Possible further practicals include:

- Investigate the structure of a mammalian heart by dissection.
- Investigate tissue water potentials using plant tissue and graded concentrations of a solute.
- Use a semi-quantitative method to estimate protein concentration using biuret reagent and colour standards.



BIOLOGY

UNIT 1-TOPIC 1A 1

THE CHEMISTRY OF LIFE

WATER

Prior knowledge

- Life process depend on molecules whose structure is related to their function.
- All living things are made up of cells.
- Many processes in living cells, including diffusion and osmosis, depend on water.
- Reactions in cells take place in solution in water.

Learning Objective

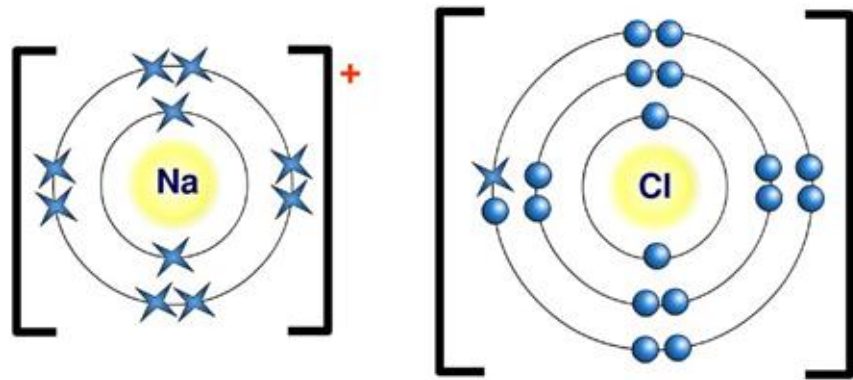
- Understand the importance of water as a solvent in transport, including its dipole nature.

K-W-L Chart

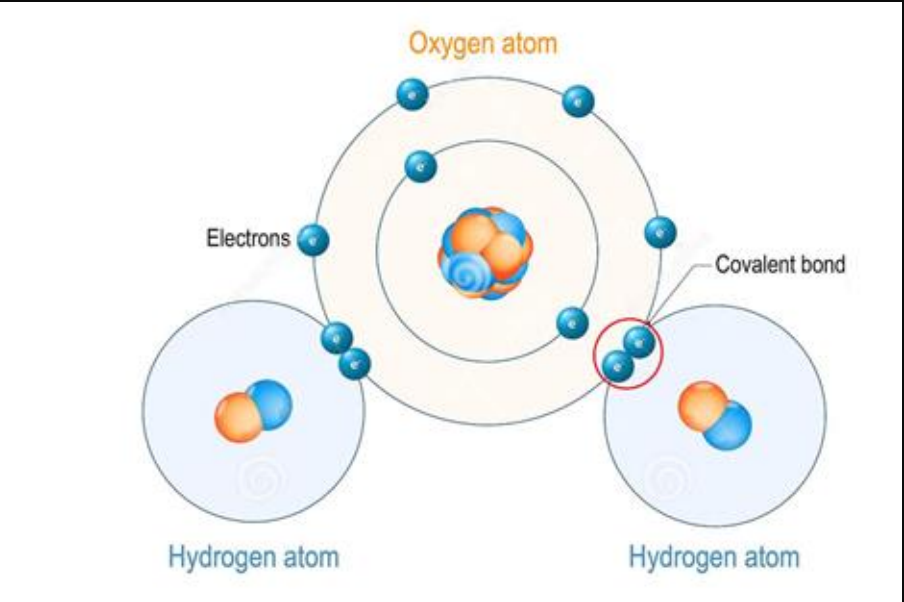
Topic: _____

What I K now	What I W ant to Know	What I L earned

Ionic bonding

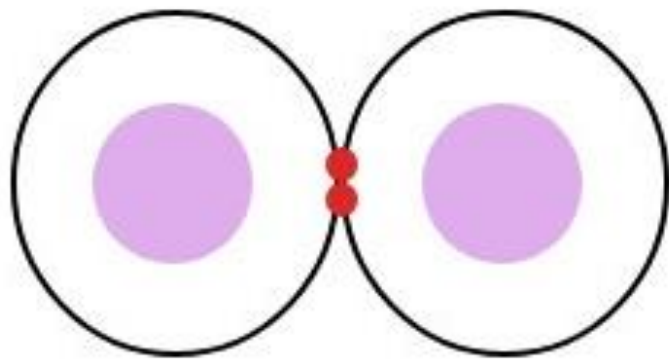
Ionic bonding	Example
• The atoms involved in the reaction give or receive electrons. • One part of the molecules gains one or more electron and becomes an <u>anion</u>(negative ion) • The other part loses one or more electrons and becomes a <u>cation</u> (a positive ion). • Ionic bonds hold the oppositely charged ions together.	The positive sodium ions and the negative chloride ions are strongly attracted to each other.  It is this electrostatic attraction that forms ionic bonds in sodium chloride and other ionic compounds.

Covalent bonding

Covalent bonding	Example
• The atoms involved in the reaction share electrons. • Covalent bonds are strong, and the molecules formed are usually neutral. • <i>In some covalent bonds, electrons are not shared evenly.</i> • So this results in a molecule with a slightly negative and slightly positive side. • This separation of charge is called <u>dipole</u>. • The molecule is described as a <u>polar molecule</u>.	

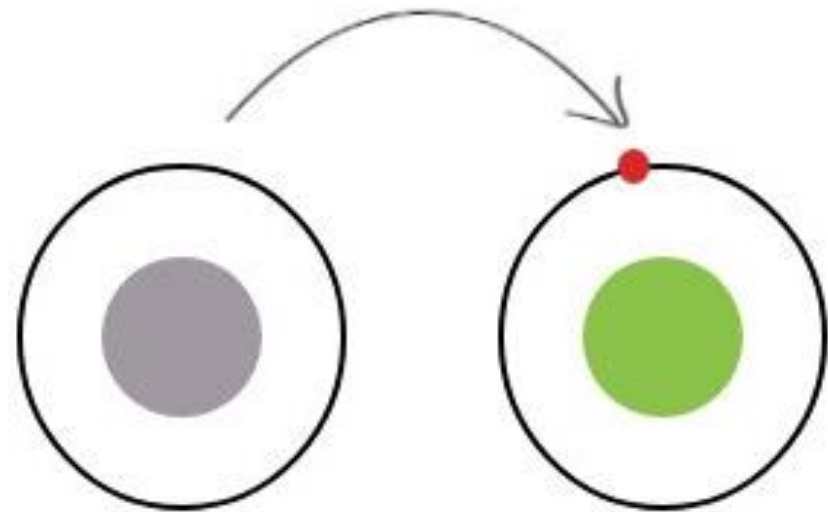
COVALENT BONDS

electrons shared



IONIC BONDS

electrons transferred



	Covalent Bonding	Ionic Bonding
Species Involved	atoms	Ions (cations and anions)
Electrons	shared	Taken by the more electronegative participant
Interactions	Balance of attractive and repulsive forces between atoms Can be polar or nonpolar depending on atoms' electronegativities	Weak electrostatic forces keep ions bonded together
Examples	H ₂ O (water) Cl ₂ (chlorine gas)	NaCl (sodium chloride) CaCO ₃ (calcium carbonate)

Electronegativity is defined as the tendency of an atom participating in a covalent bond to attract the bonding electrons.

INORGANIC IONS

- Occur in solution in the cytoplasm and body fluids of organisms
- Some in high concentrations and others in very low concentrations
- Each type of ion has a specific role, depending on its properties and these roles are relevant in a range of topics across the A Level
- Ionic substances dissolve in water, the ions separate in a process called **dissociation**.

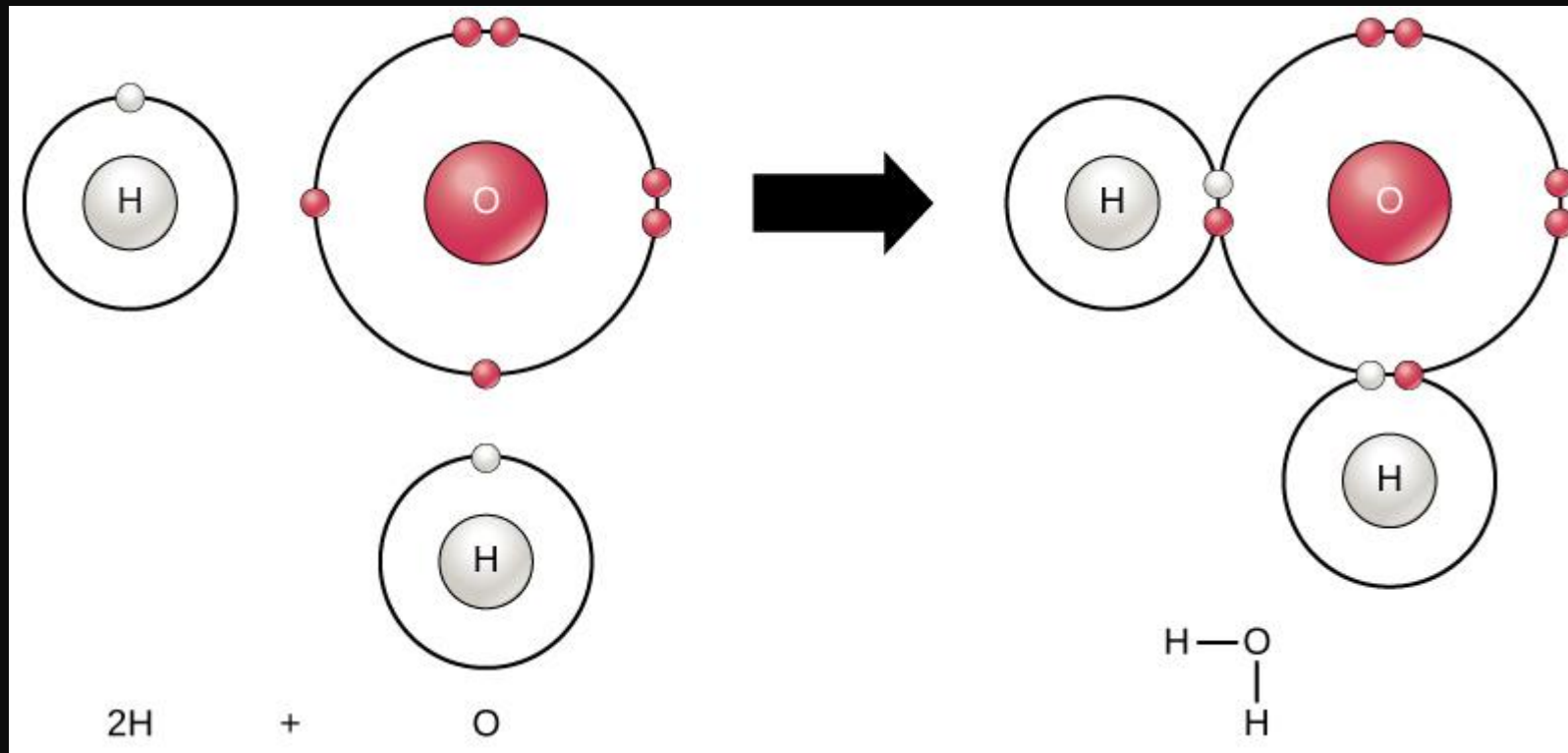
Ion	Chemical symbol	Role
Phosphate	PO_4^{3-}	Attached to other molecules as a phosphate group, for example: • in DNA nucleotides, enabling nucleotides to join together forming phosphodiester bonds • in ATP – bonds between these store / release energy
Hydrogen	H^+	Maintain pH levels in the body • Too much H^+ = acidic (low pH) • Too little H^+ = alkaline (high pH) • Affects rate of enzyme-controlled reactions as can cause enzymes to denature
Iron	Fe^{2+}	• Component of (haem group of) haemoglobin which is contained in red blood cells • Transports oxygen around the body – oxygen temporarily binds to it, so it becomes Fe^{3+}
Sodium	Na^+	• Co transport of glucose and amino acids across cell membranes • Involved in generating nerve impulses and muscle contraction (year 2)

WATER



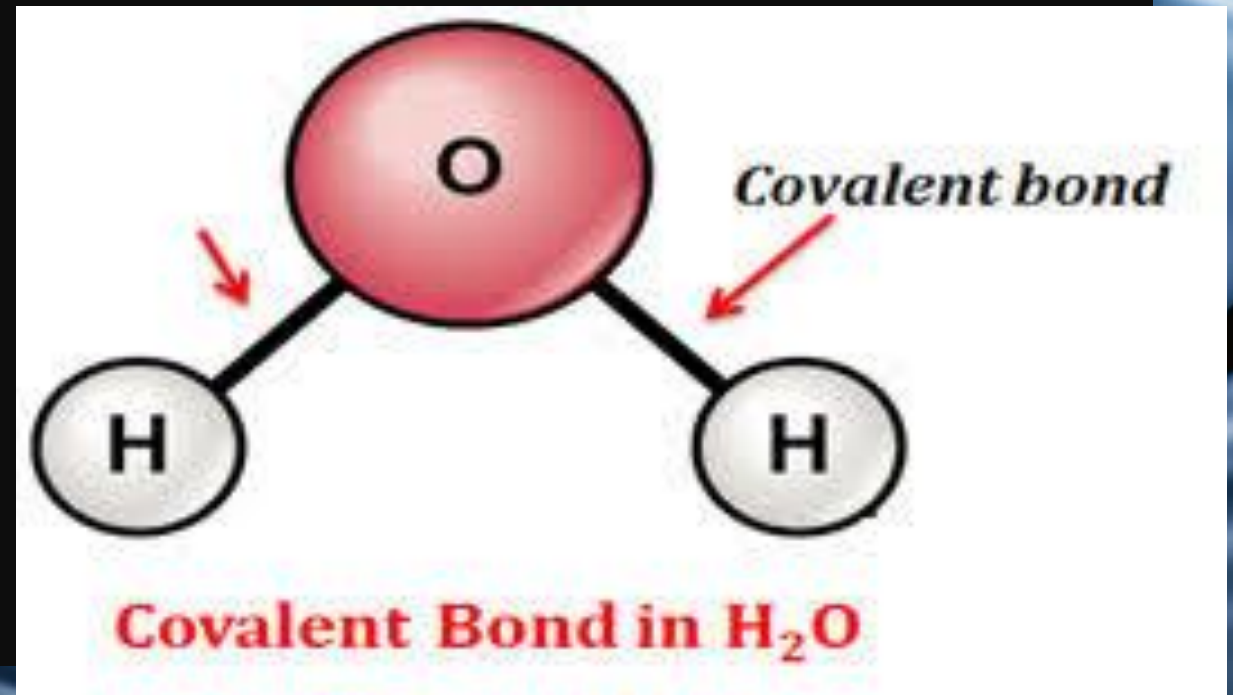
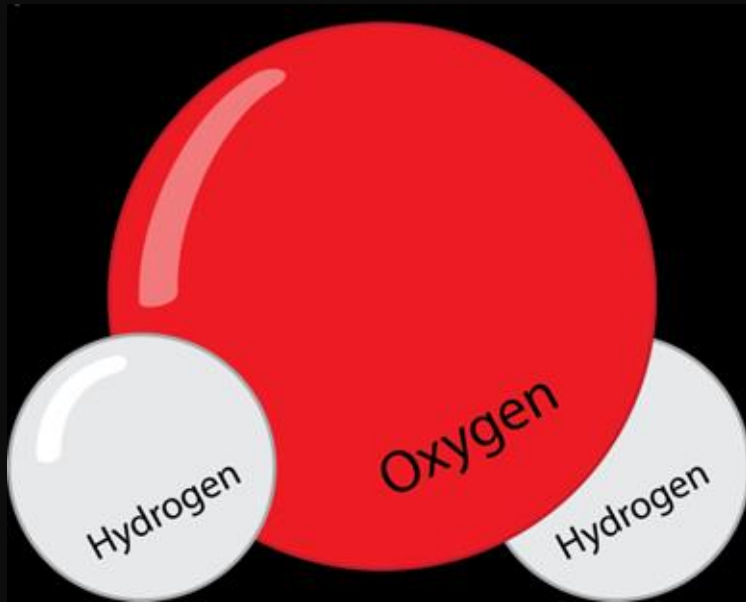
- Water exists in different physical states – solid, liquid and gas.
- Makes up 70% of the surface of Earth.
- 65 – 90% of the weight of all living organisms.
- Plays important role in all vital processes of living organisms.
- Chemical formula is H_2O .

Water molecule

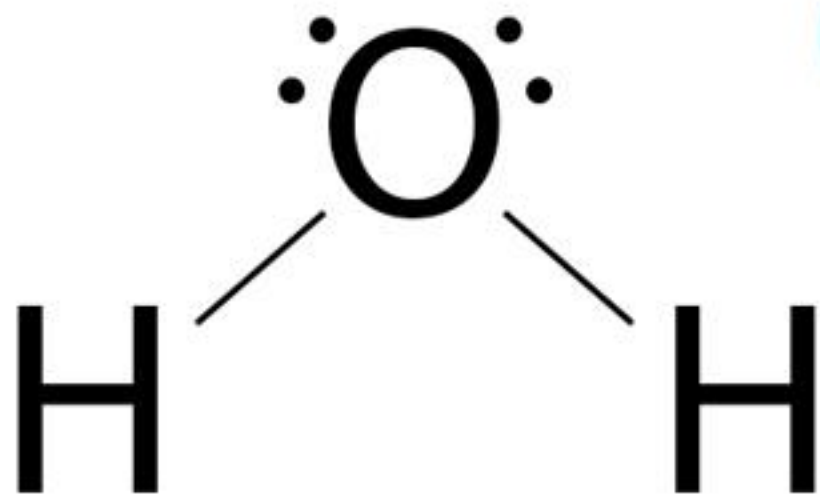


Structure

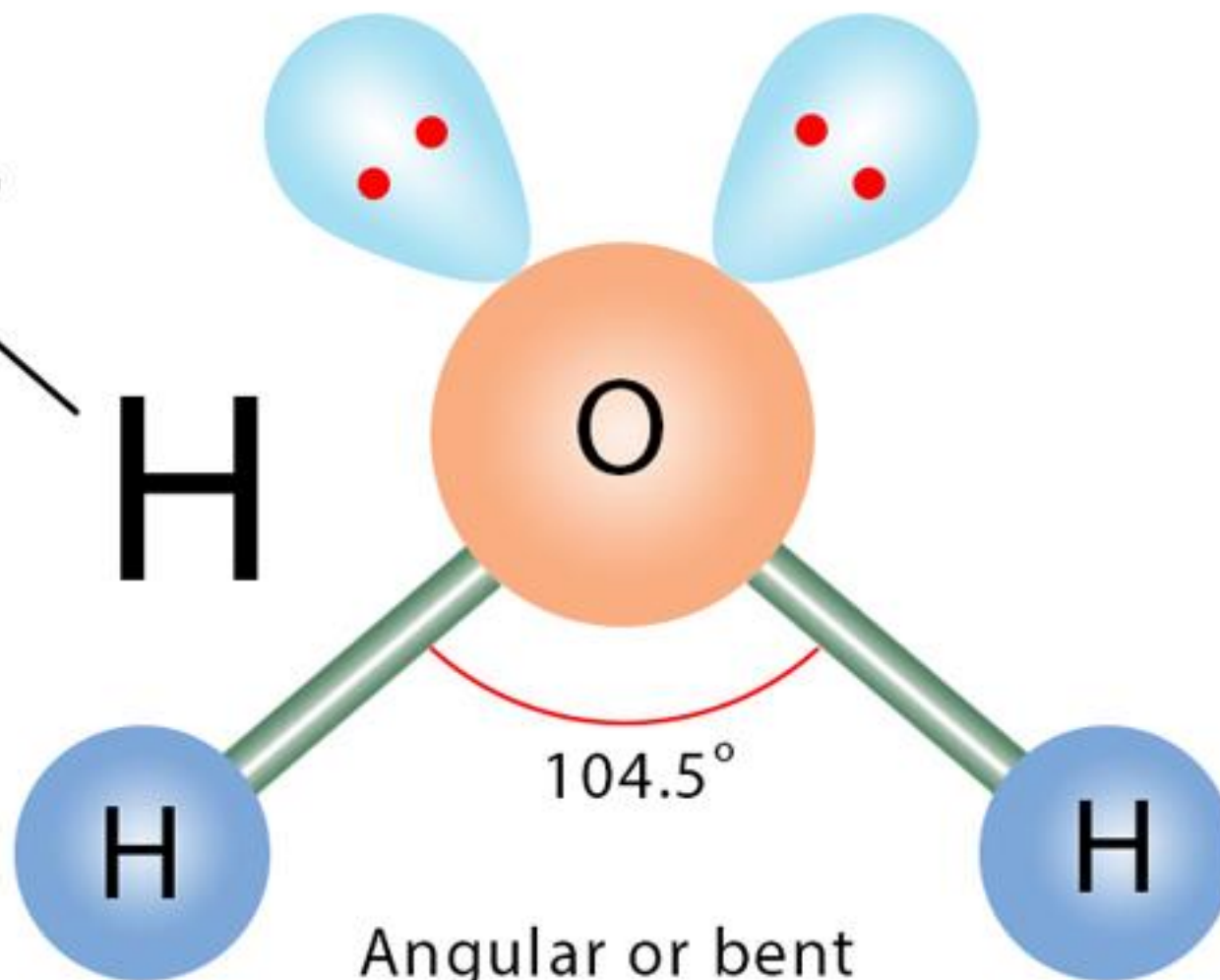
- A water molecule consists of two Hydrogen atoms covalently bonded to an Oxygen atom.



Molecular Geometry of Water (H_2O)



Because lone pairs push bonding pairs closer together, the H–O–H angle decreases to **about 104.5°** .



Water molecule

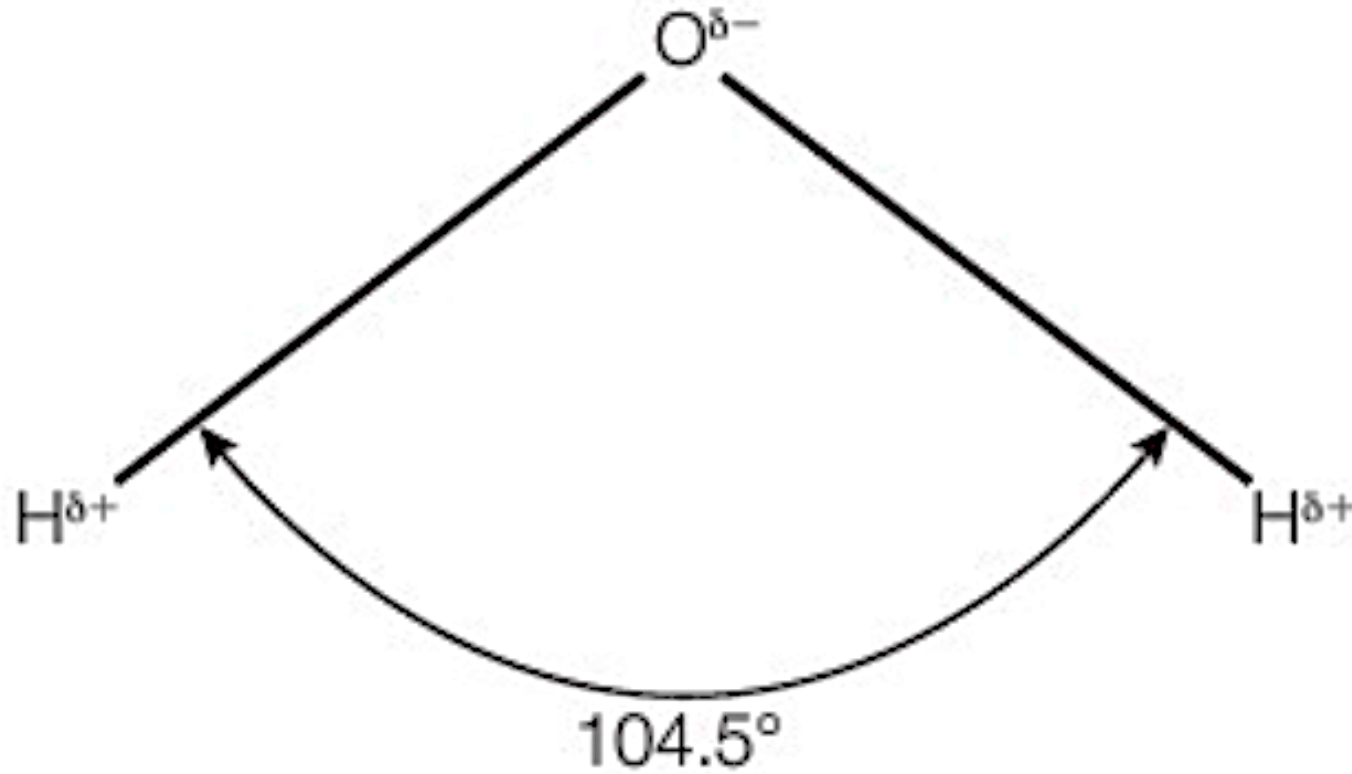


fig F

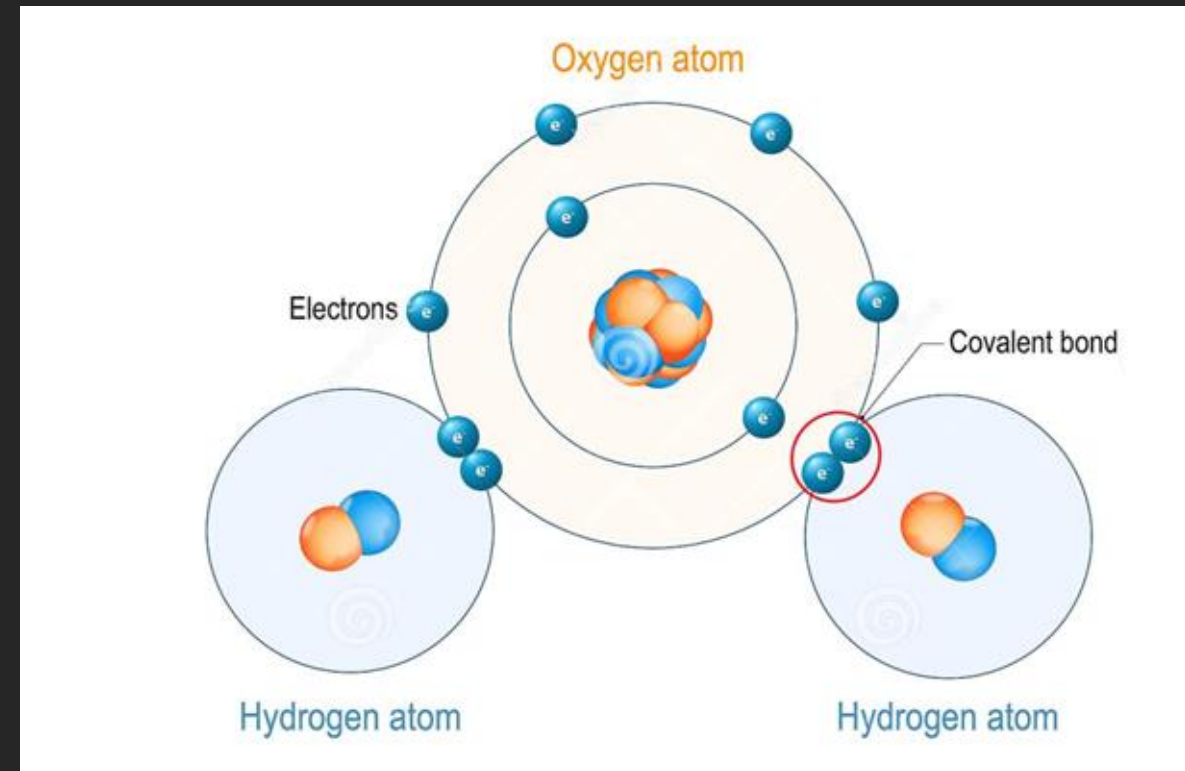
A model of a water molecule showing dipoles.

Dipole nature of the water molecule

Covalent bonding

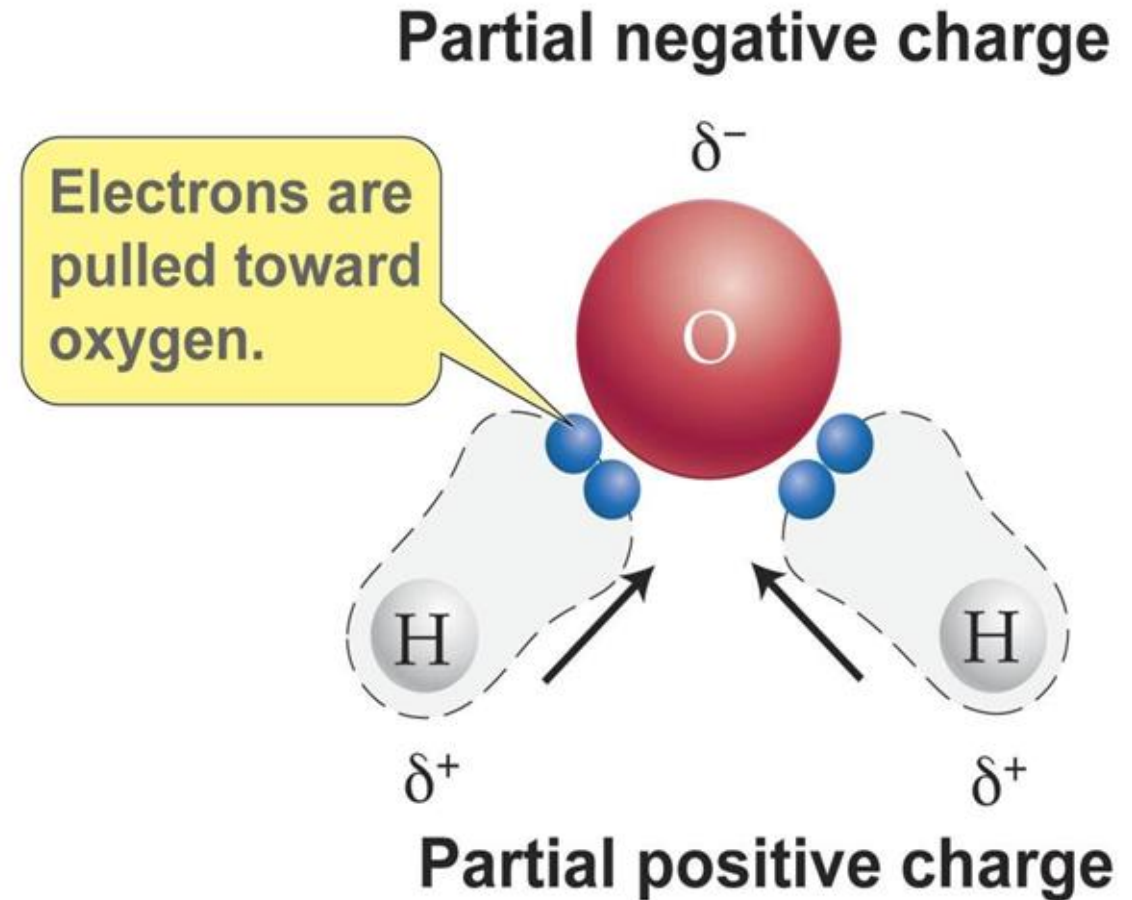
- In water molecules, oxygen and hydrogen form covalent bonds.
- These atoms share electrons.
- Covalent bonds are strong and the molecules formed are usually neutral.
- Some covalent bonds are not shared evenly.
- So this results in a molecule with a slightly negative and slightly positive part.
- This separation of charge is called dipole.
- The molecule is describe as a polar molecule.

Example

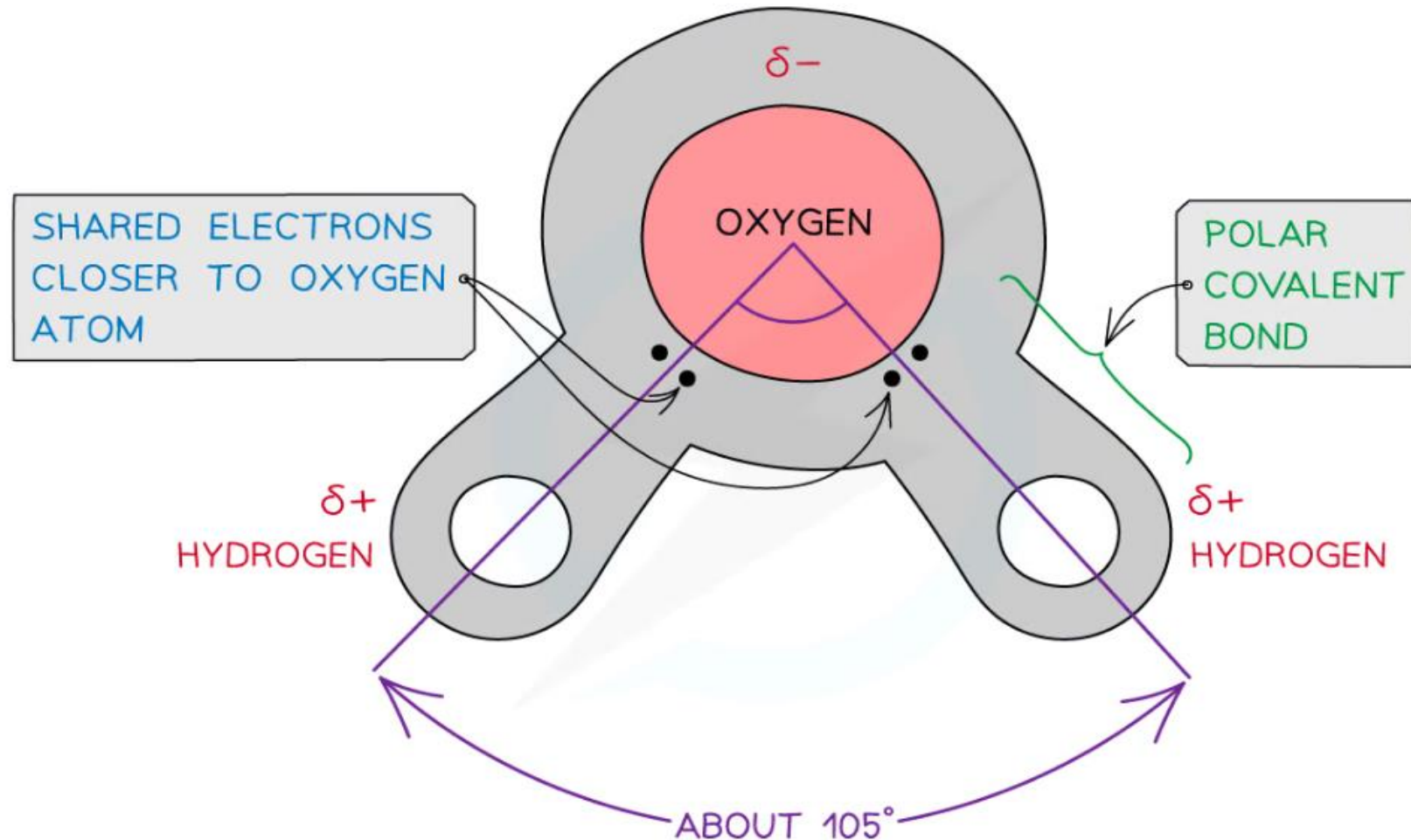


Dipole nature of the water molecule

- The hydrogen end of the molecule is slightly positive.
- The oxygen end is slightly negative because the electrons are more concentrated at that end.
- Water is therefore called a **Polar molecule**.
- It is this polarity that accounts for many of its biologically important properties.

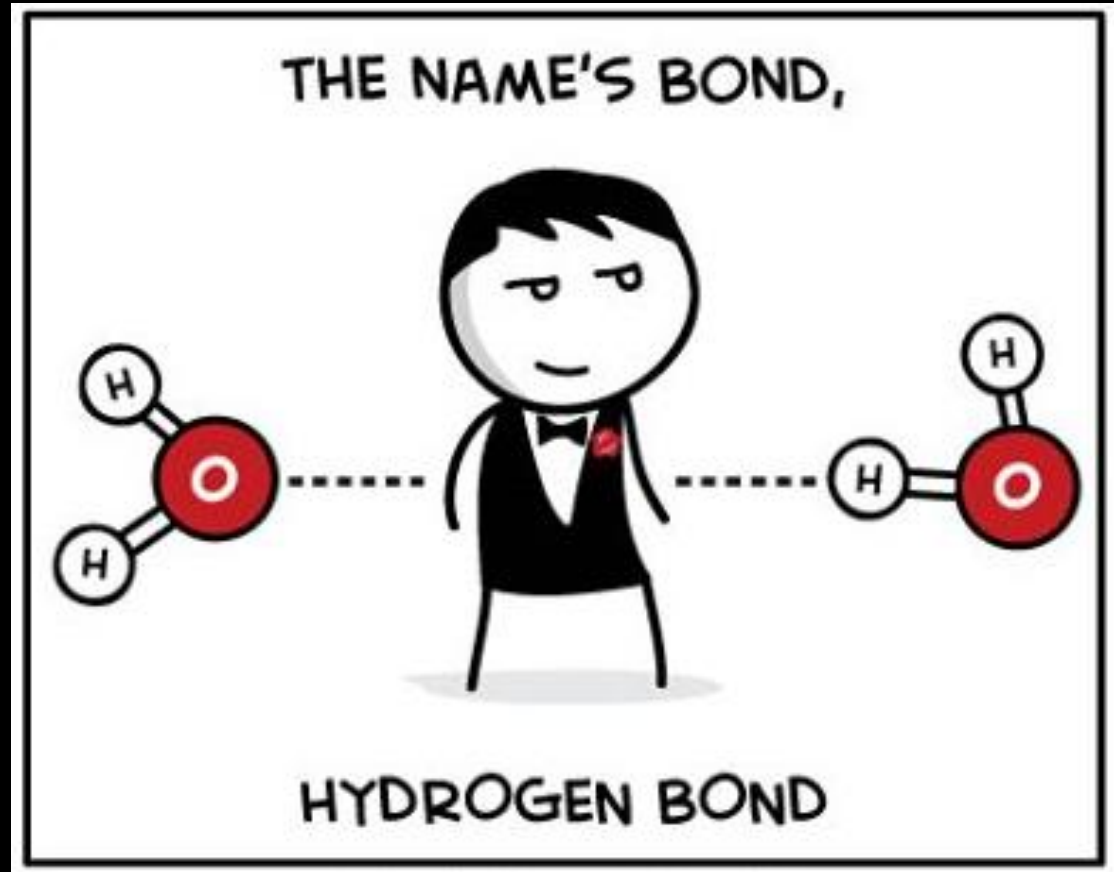


Dipole nature of the water molecule



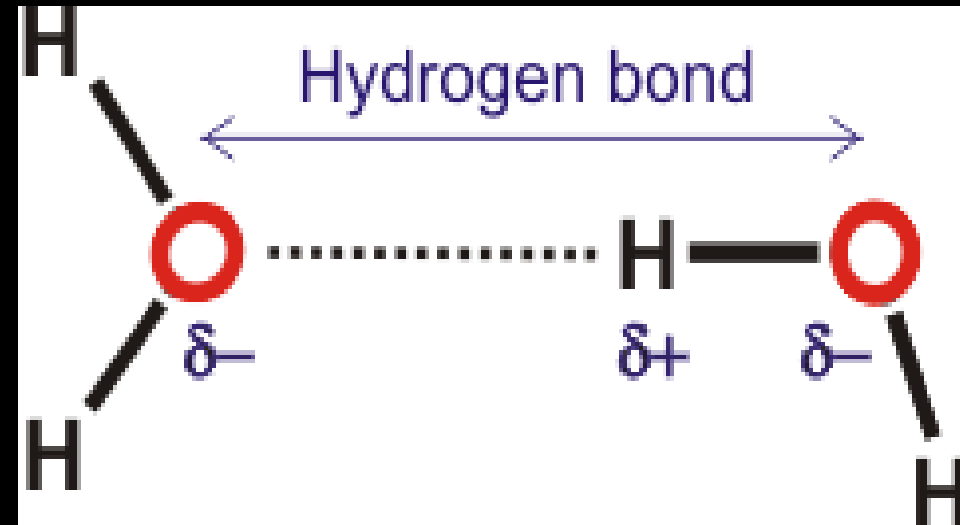
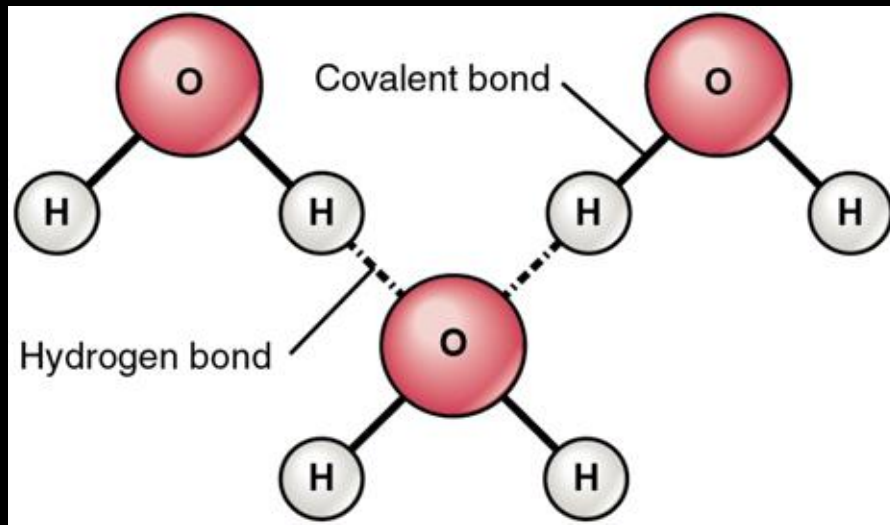
Dipole nature of the water molecule

➤ One major effect of polarity is water molecules form hydrogen bonds.

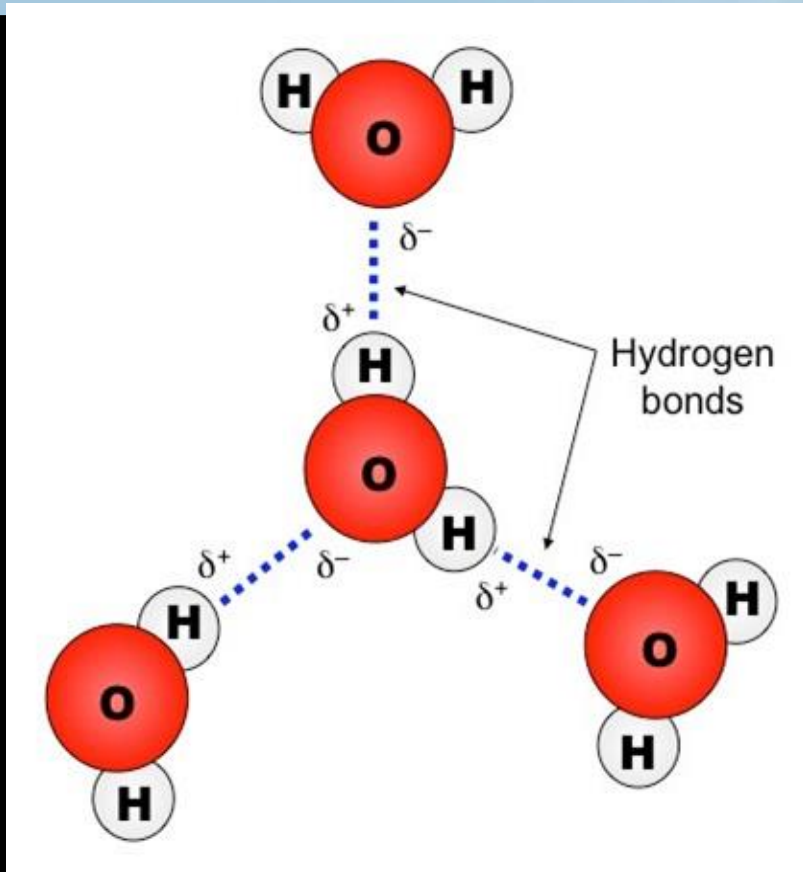


Dipole nature of the water molecule

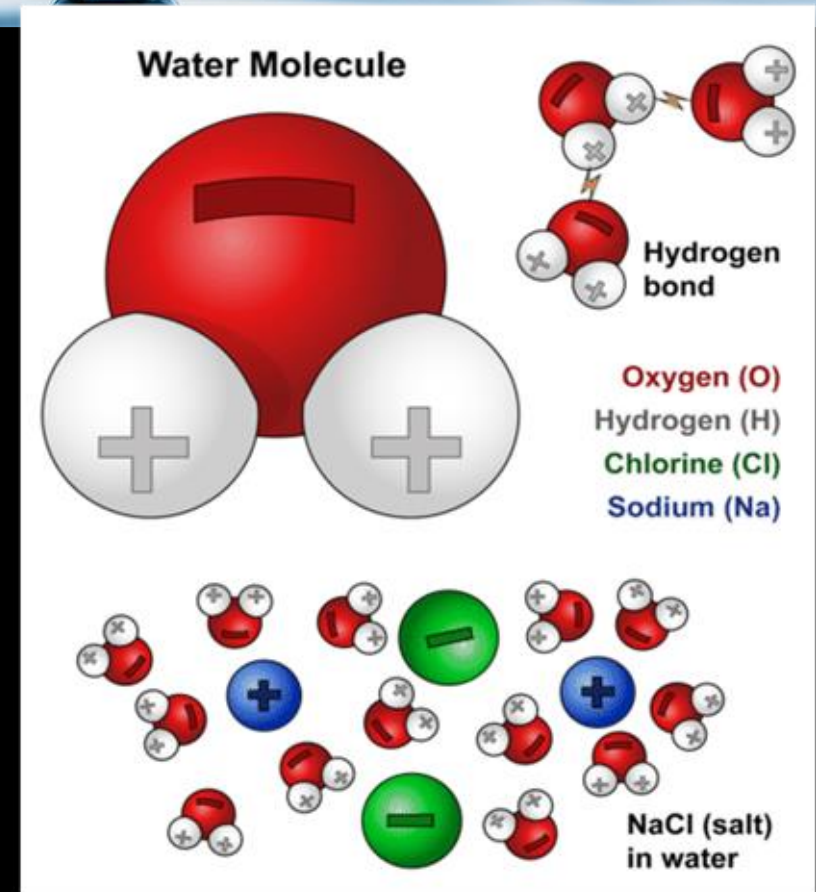
- The charge differences cause water molecules to be attracted to each other and to other polar molecules and ionic substances.
- Each individual hydrogen bond is weak but there are many of them so the molecules of water stick together.



Dipole nature of the water molecule



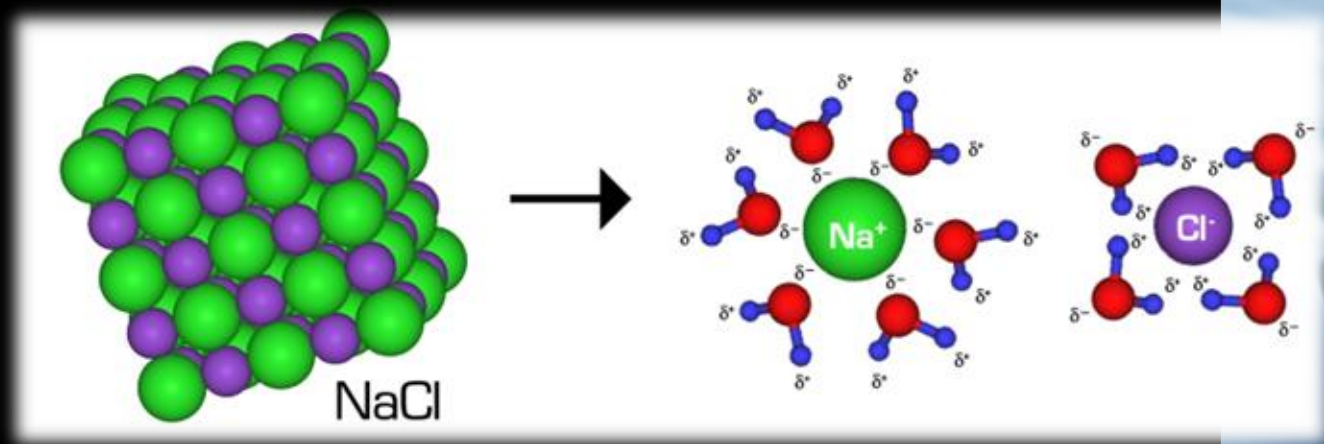
(a) Water to water attraction



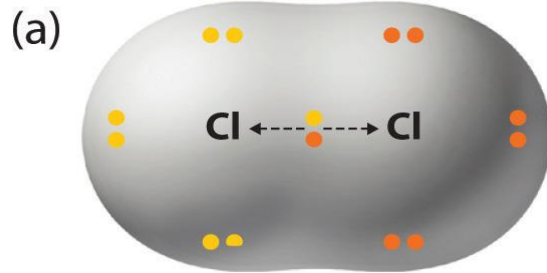
(b) Water to other molecules attraction

Properties of water: POLAR SOLVENT

- Water is a polar solvent.
- Many ionic substances will dissolve in it.
- Many covalently bonded substances are also polar and will dissolve in water.
- Such polar substances are said to be hydrophilic – ‘water-loving’.

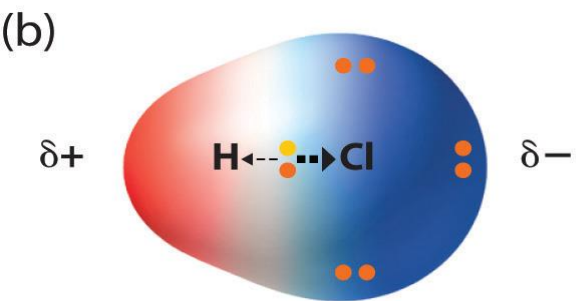


It is important to note that polar molecules are not the same as ionic molecules!!



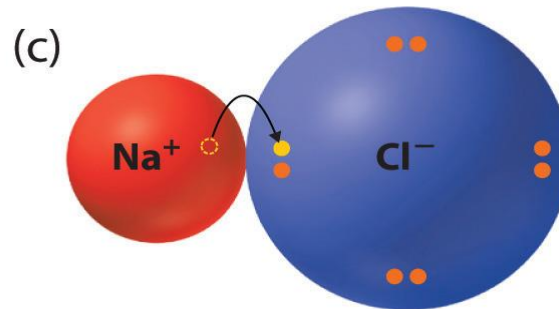
Nonpolar covalent bond

Bonding electrons shared equally between two atoms.
No charges on atoms.



Polar covalent bond

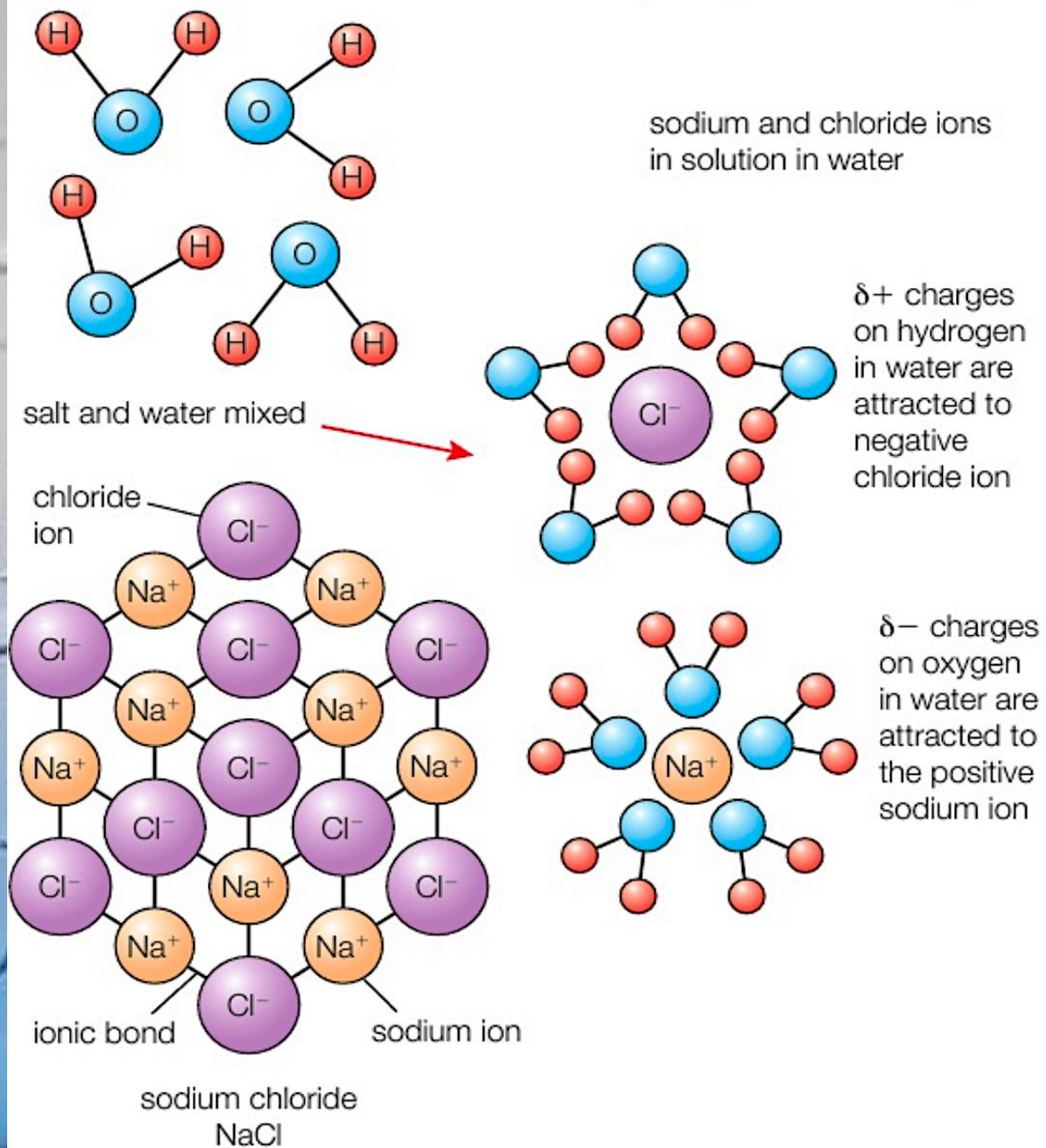
Bonding electrons shared unequally between two atoms.
Partial charges on atoms.



Ionic bond

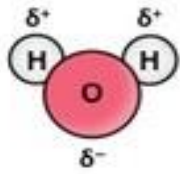
Complete transfer of one or more valence electrons.
Full charges on resulting ions.

1. Ionic compounds are held together by ionic bonds that arise due to electrostatic forces between ions. In such cases, one of the atoms loses an electron to form a cation and another atom gains an atom to form an anion. Eg: Na⁺/Cl⁻
2. In polar compounds the electron pair is shared by two chemical species. However, the electron pair is shared in an unequal manner owing to the differences in the electronegativities of the two chemically bonded species. This results in one side of molecule to have a partially positive charge while the other side have a partially negative charge. Eg: glucose/amino acids/molecules with OH/NH₂ groups.



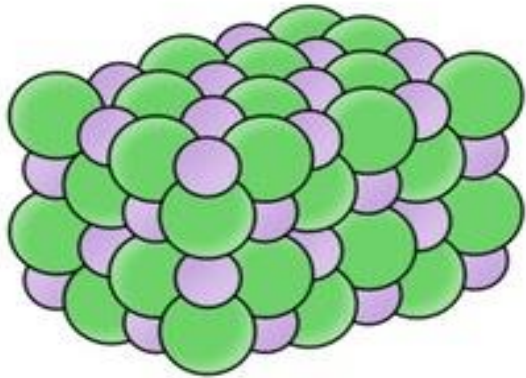
▲ **fig H** A model of sodium chloride dissolving in water as a result of the interactions between the charges on sodium and chloride ions and the dipoles of the water molecules.

- ❑ **Water is a polar solvent and ions are charged.**
- ❑ Water separates the ions from the lattice.
- ❑ Water surrounds/ clusters around the ions.
- ❑ **Sodium ions will bind to / be attracted to the oxygen of water.**
- ❑ **And chloride ions will bind to / be attracted to the hydrogens of water.**
- ❑ This way, Na^+ and Cl^- ions dissolves in water.

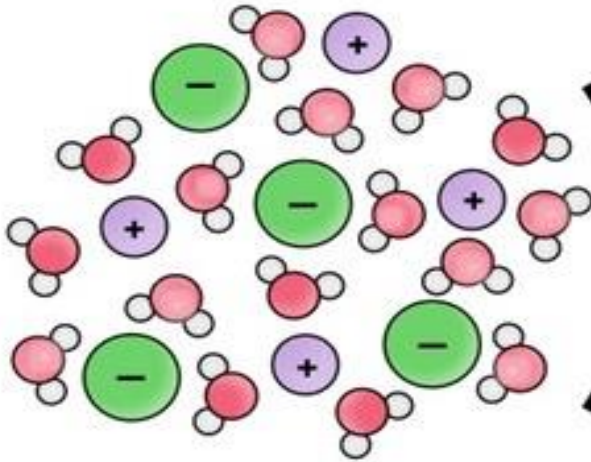


Water Molecule (H₂O)

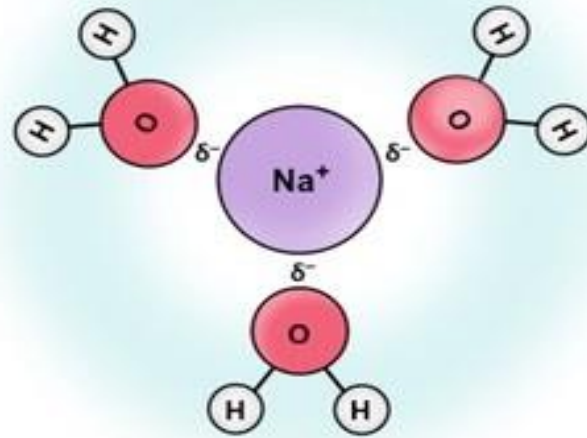
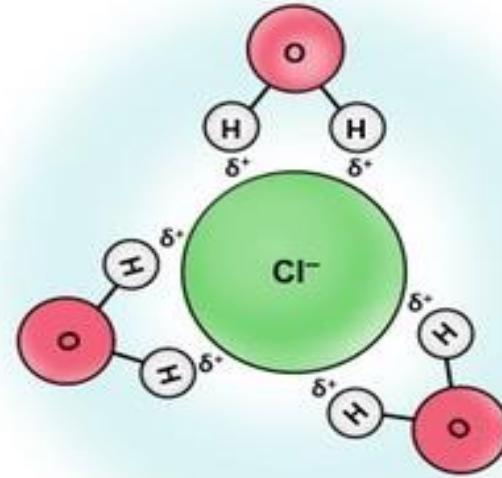
+

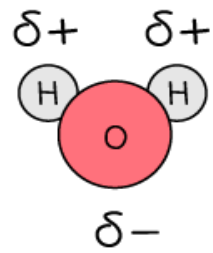


Salt Crystal (NaCl)

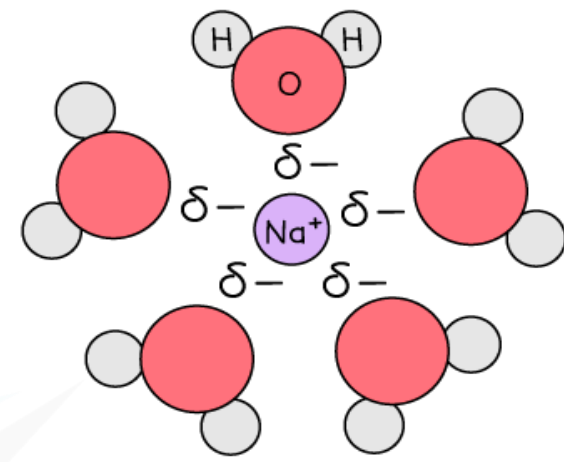


Solution

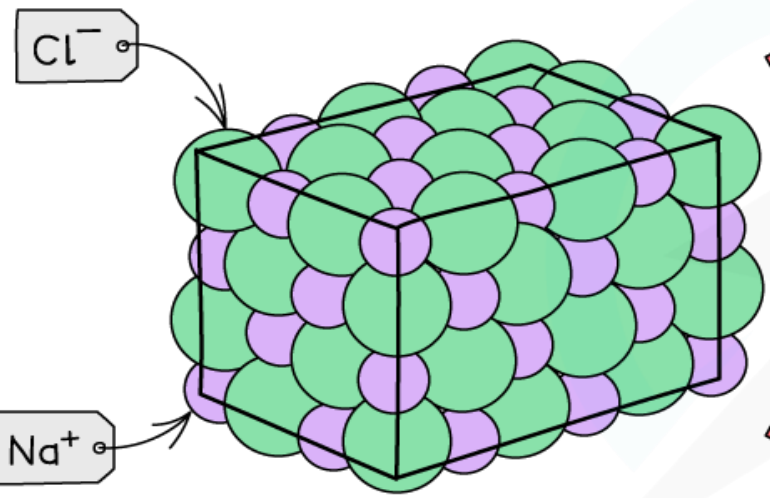




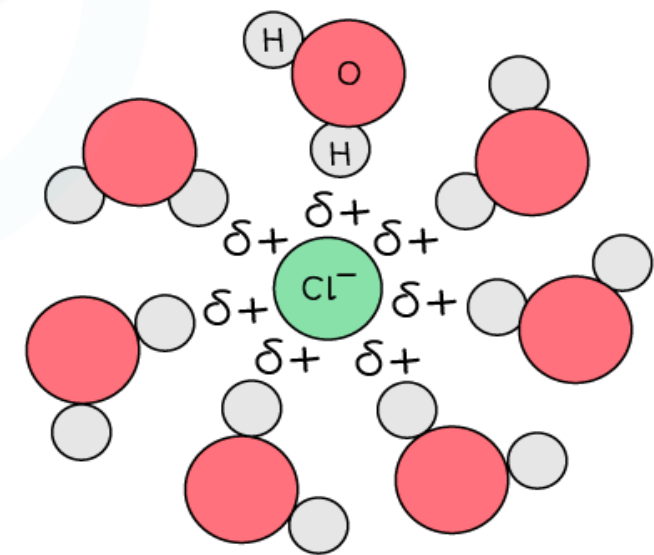
WATER MOLECULE



HYDRATED SODIUM ION

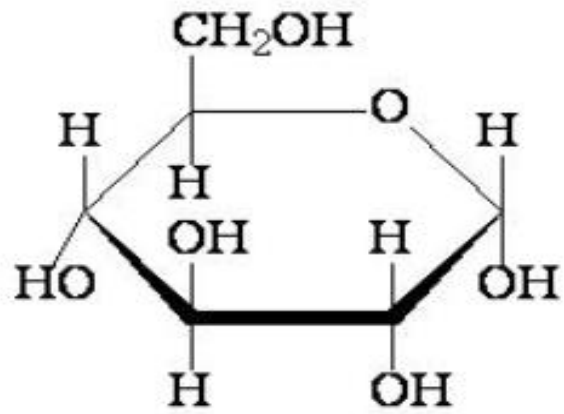


CRYSTAL OF NaCl



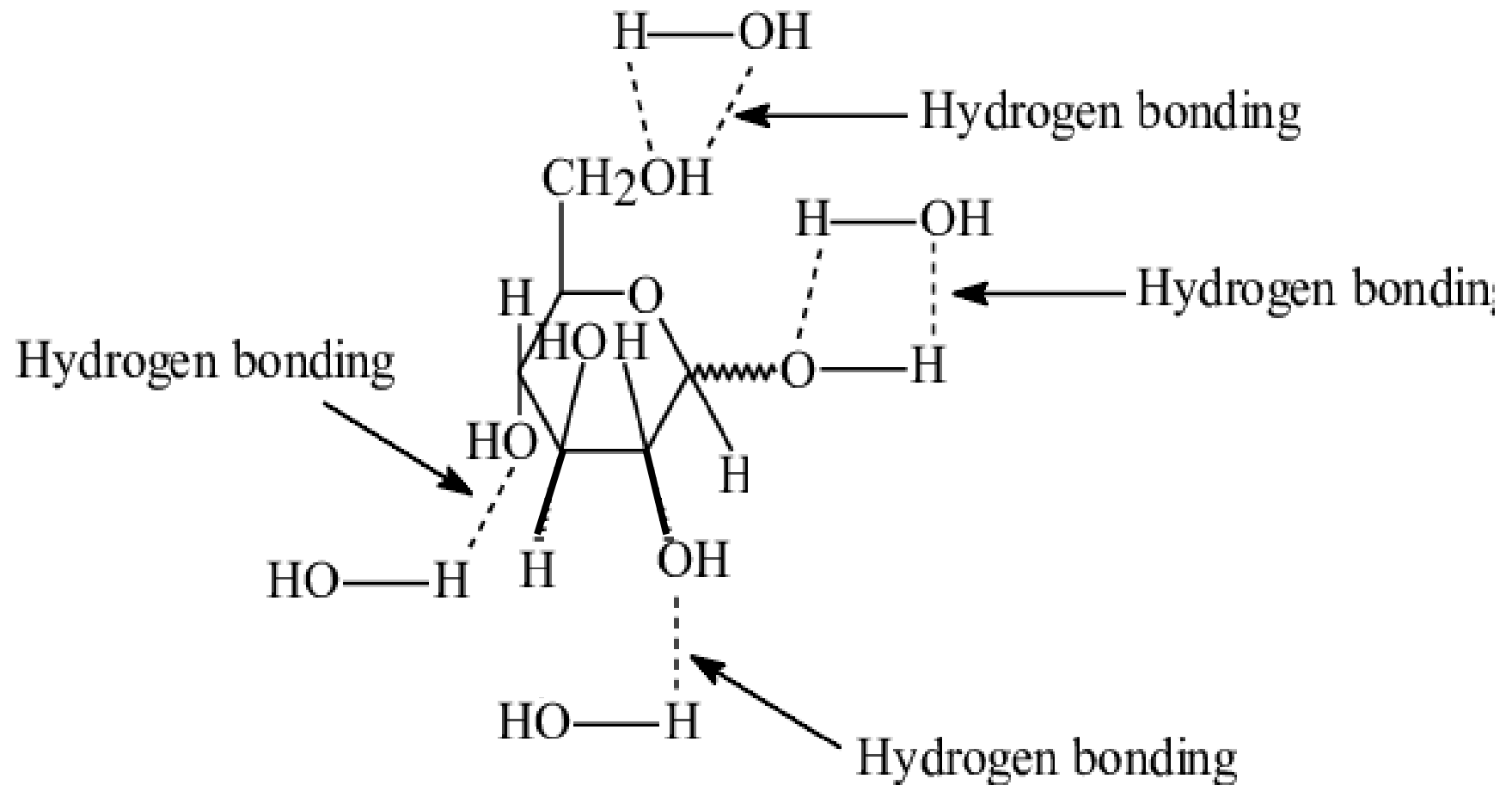
HYDRATED CHLORIDE ION

- ❑ The surrounded molecules break apart e.g. sodium chloride molecules break into sodium ions and chloride ions
- ❑ The ions surrounded by water molecules have dissolved.
- ❑ This allows chemical reactions to occur within cells as the dissolved solutes are more chemically reactive when they are free to move about.
- ❑ Metabolites can be transported efficiently in a dissolved state.

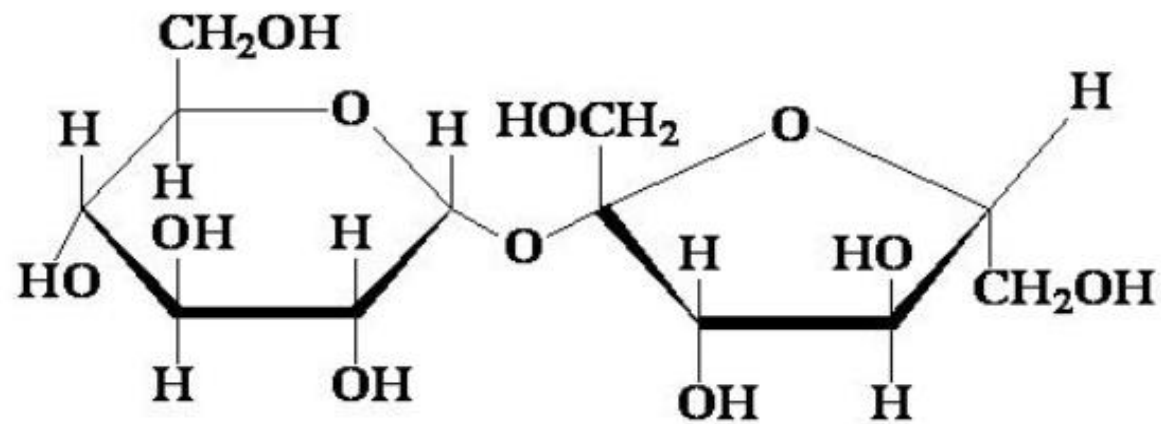


glucose

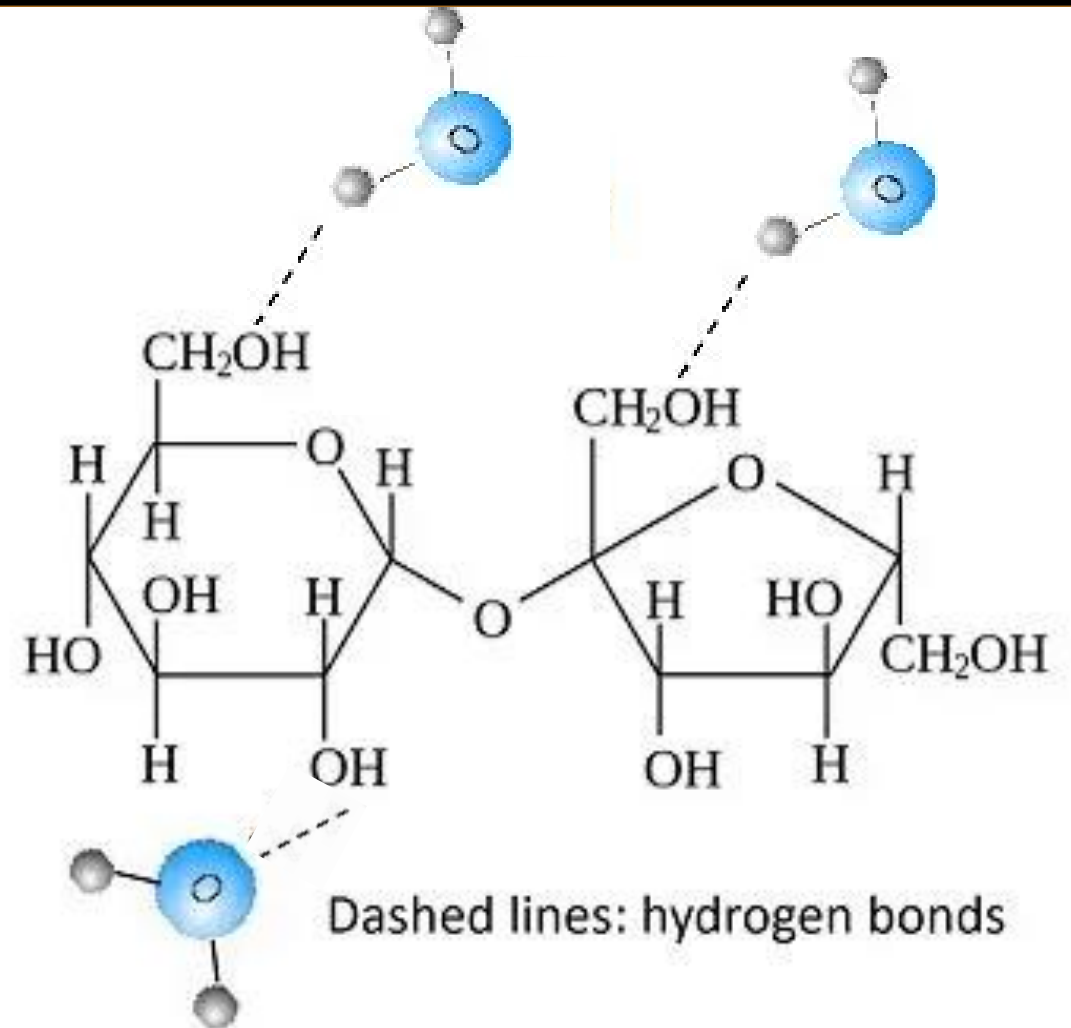
□ Hydroxyl (OH) groups in glucose forms **hydrogen bonds** with water and dissolves in water.



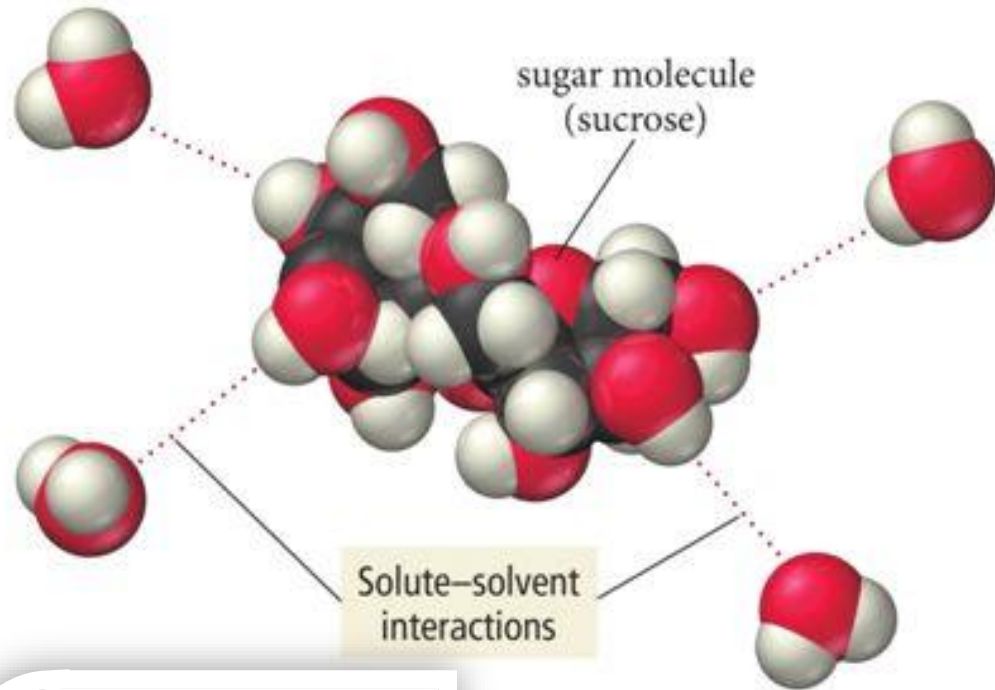
□ Hydroxyl (OH) groups in sucrose forms **hydrogen bonds** with water and dissolves in water.



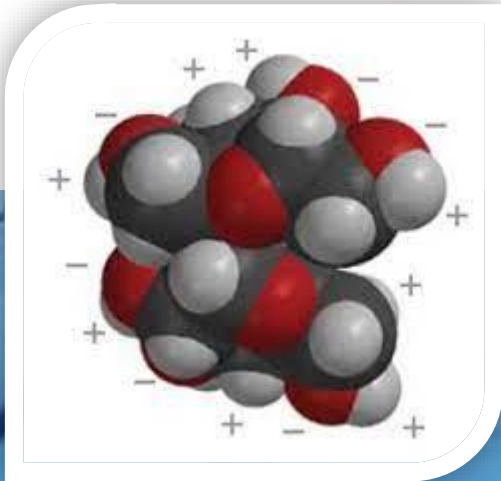
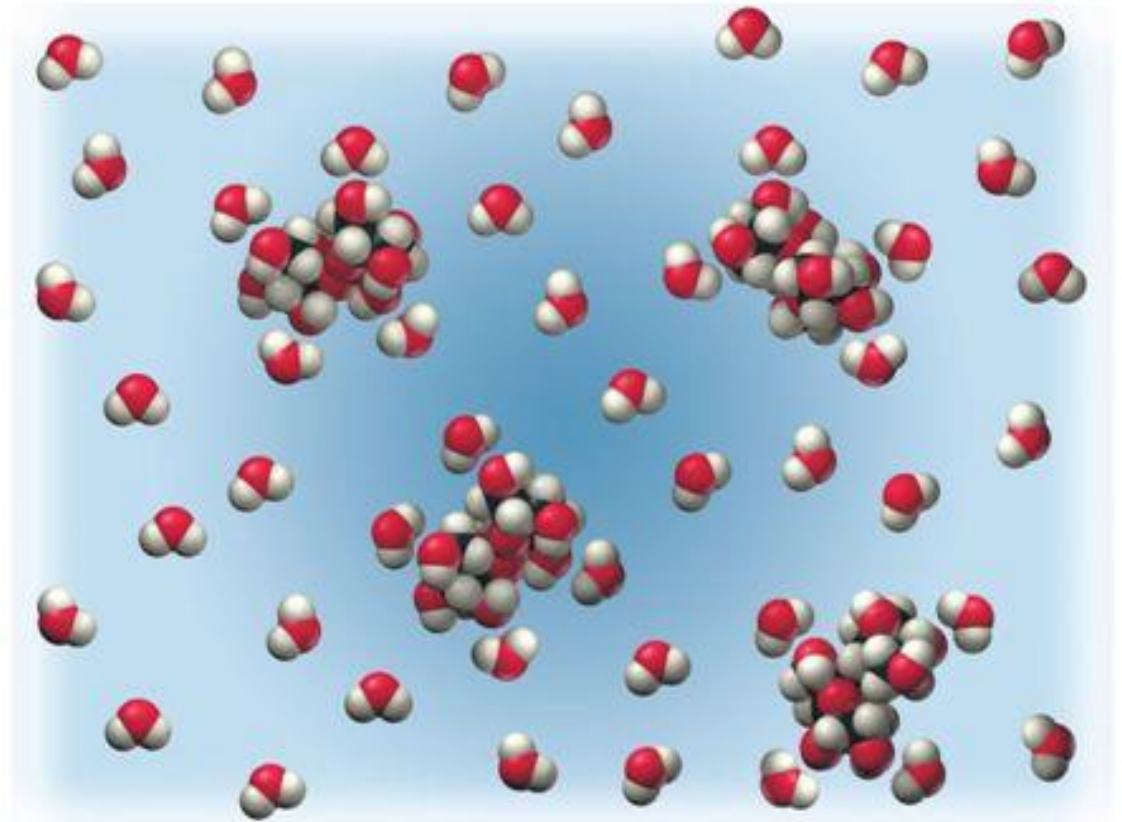
sucrose



Interactions between Sugar and Water Molecules



Sugar Solution

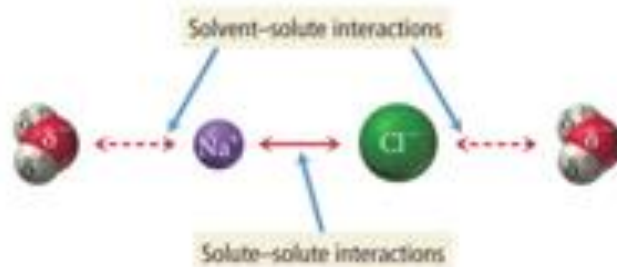


Polar sugar molecule

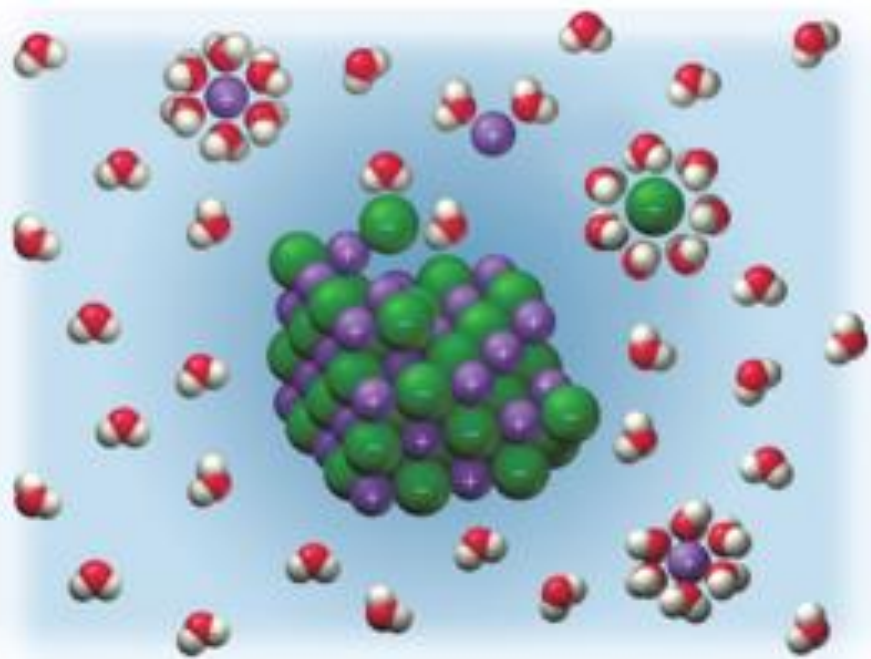
- ☐ Sugars such as glucose/sucrose are larger than NaCl, hence can be surrounded by more water molecules.
- ☐ Sugars have more hydroxyl groups/ polar groups, therefore they are able to form more H-bonds with water molecules.
- ☐ Therefore they are more soluble in water than ionic compounds like NaCl.

NaCl solution

Interactions in a Sodium Chloride Solution

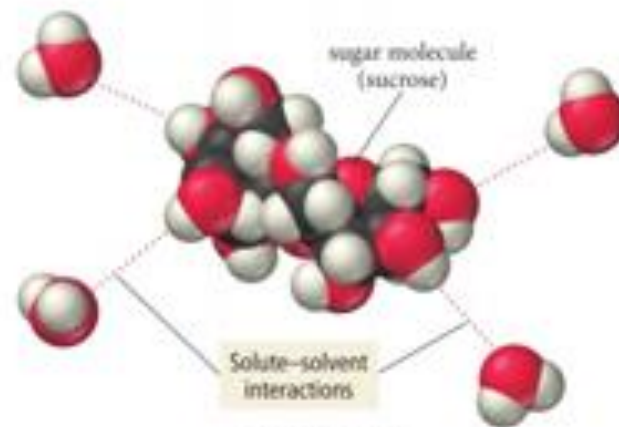


Dissolution of an Ionic Compound

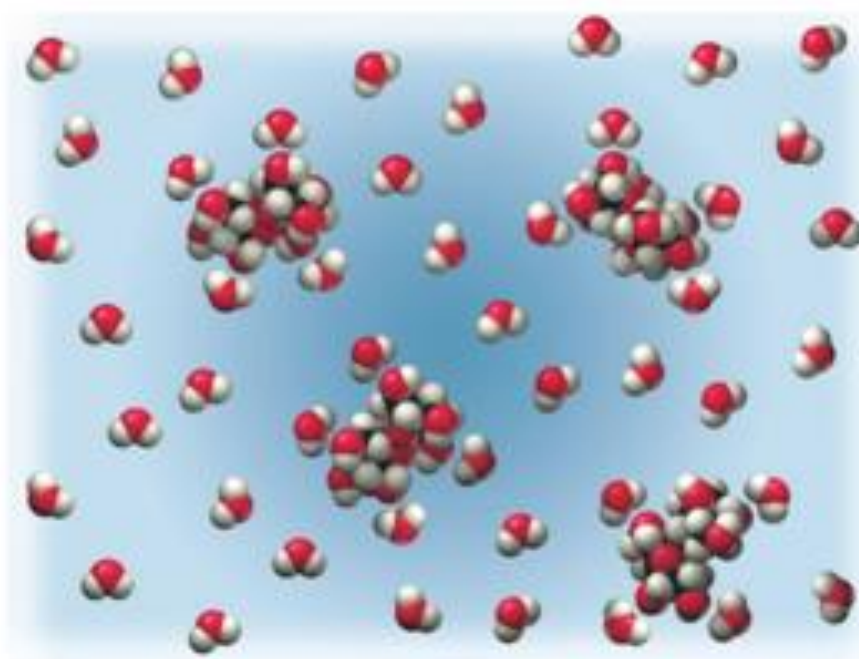


Sugar-water solution

Interactions between Sugar and Water Molecules



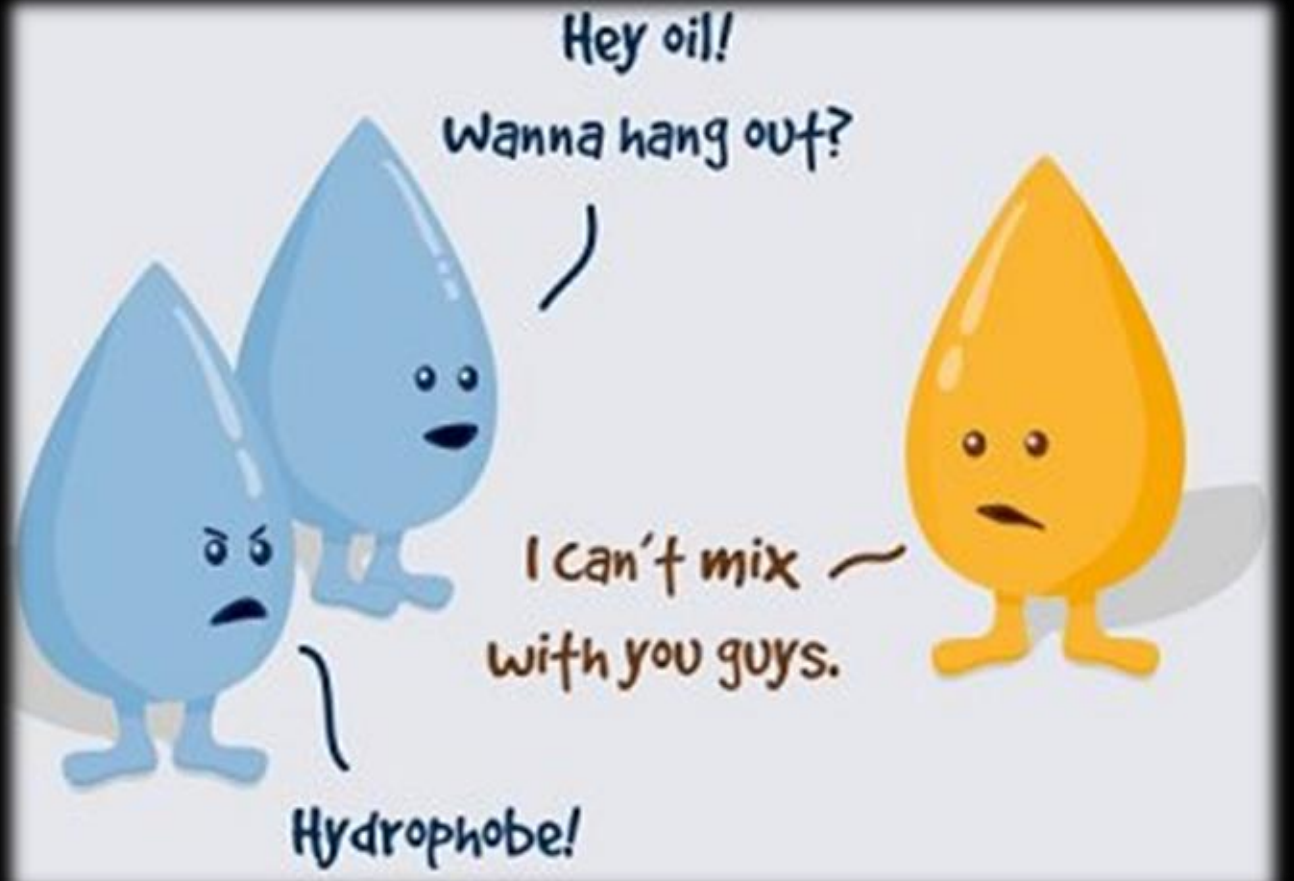
Sugar Solution



Properties of water:

POLAR SOLVENT

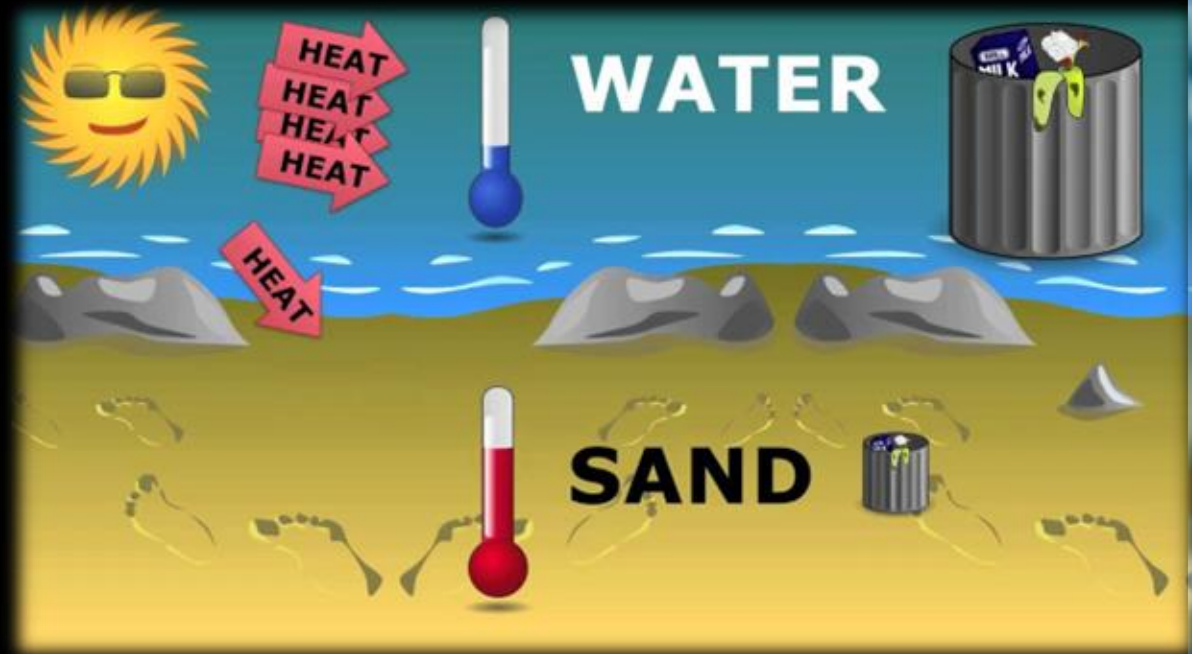
- Non-polar, hydrophobic - "water hating" substances, such as lipids, do not dissolve in water.
- Fatty acids have hydrophobic tails, they do not form H-bonds with water.
- Water also carries substances such as starch.



Properties of water:

HIGH SPECIFIC HEAT CAPACITY

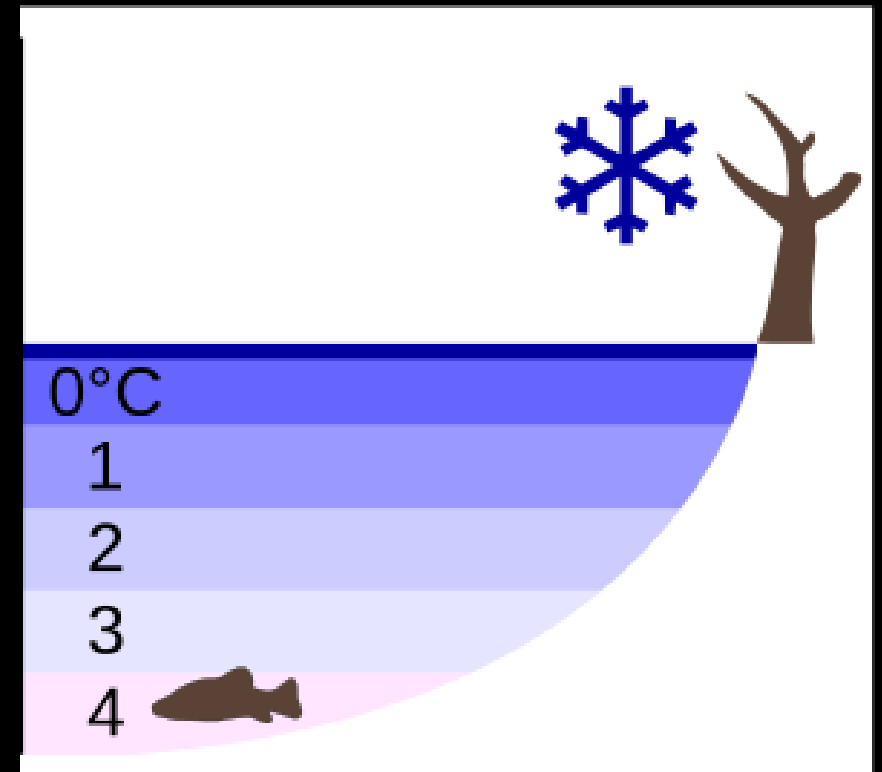
- Water is slow to absorb and release heat, hence it has a high specific heat capacity.
- The presence of many hydrogen bonds between the water molecules means that a lot of heat energy is needed to separate the water molecules from each other.
- So, the temperature of large bodies of water/ lakes/ rivers/ oceans remains fairly stable. This makes them good habitats for organisms.



Properties of water:

DENSITY DECREASES UNDER 4°C

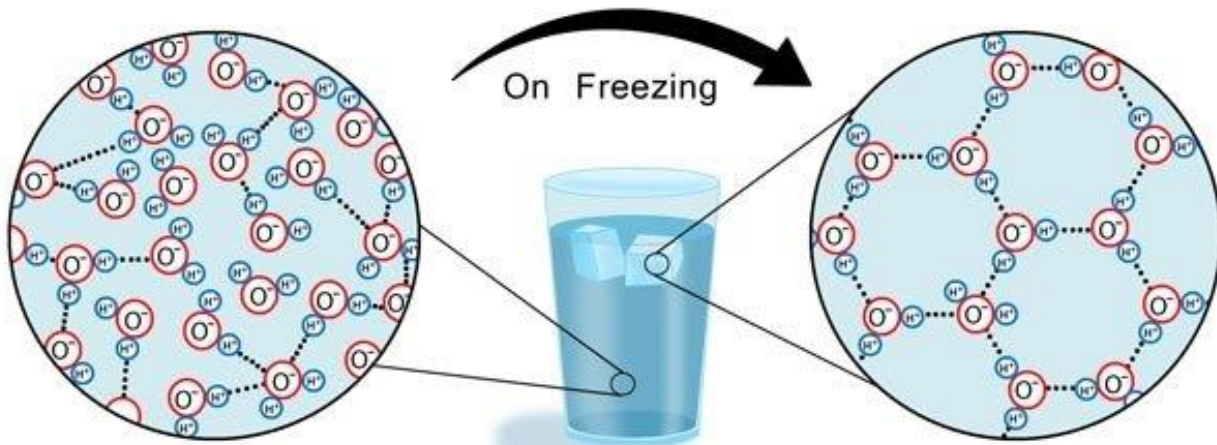
- As water cools to 4°C, it reaches its maximum density.
- Ice is less dense than water and floats, forming an insulating layer.
- This property enables organisms to live in water even in very cold winters which can otherwise result in freezing of water bodies/habitats.



Properties of water:

DENSITY DECREASES UNDER 4°C

WHY DOES ICE FLOAT ON WATER



In liquid state, water molecules are closely held together by weak hydrogen bonds

On reaching solid state, hydrogen bonds become stable, arranging the water molecules far apart from each other

This causes ice to have a lesser density than water, making it float

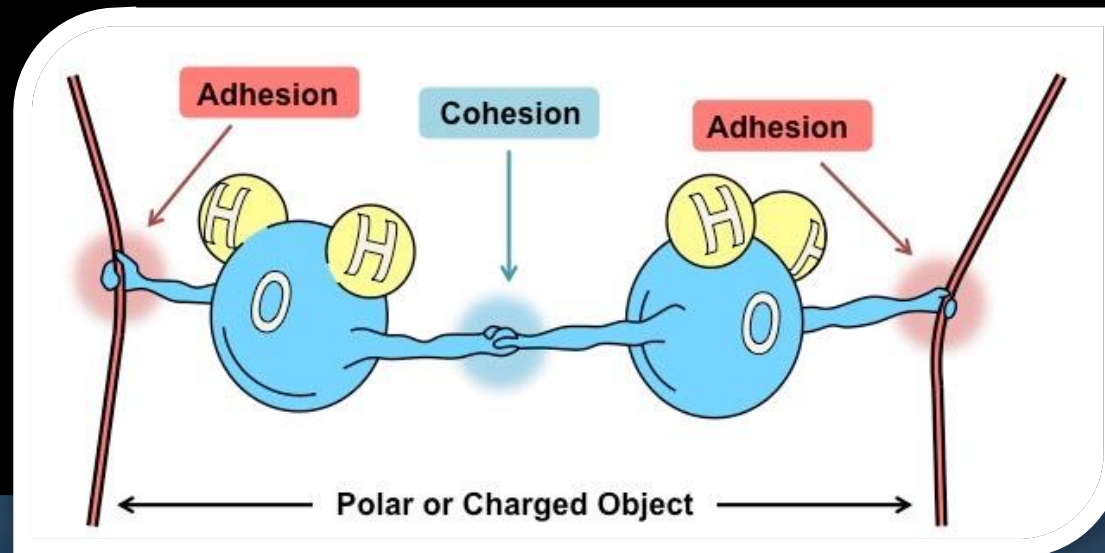
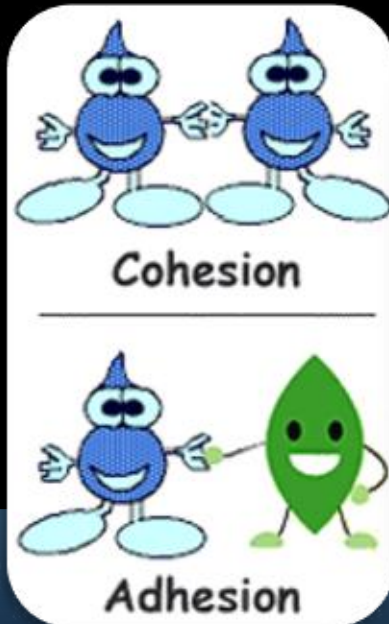


- Ice provides habitat for organisms such as penguins in Antarctica.
- And water underneath is insulated by the ice protecting and providing a habitat for aquatic organisms.

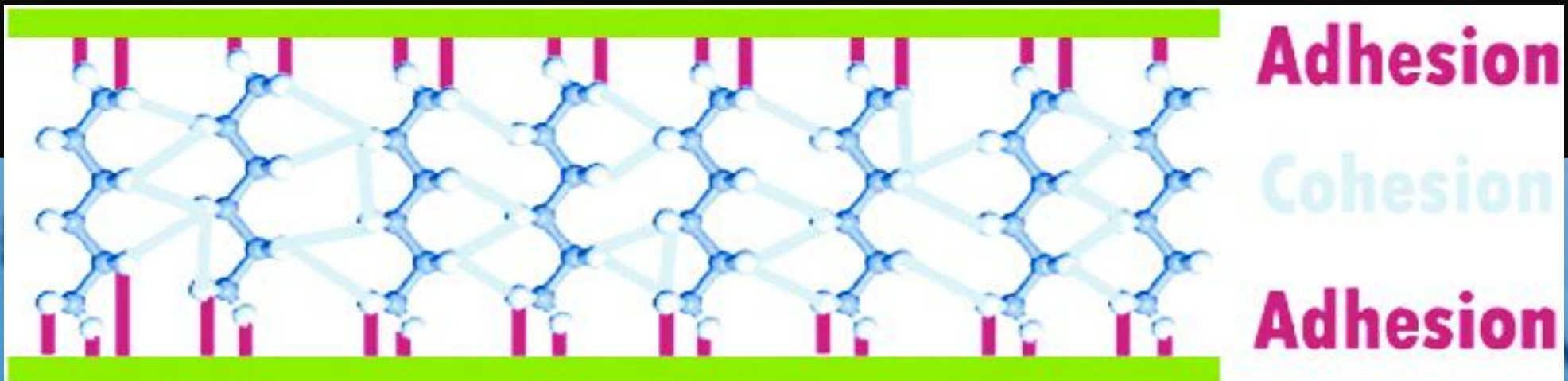
Properties of water:

COHESION AND ADHESION

- Water molecules are **cohesive**-the force between the water molecules mean they stick together.
- Water molecules are **adhesive**- they are attracted to other different molecules.



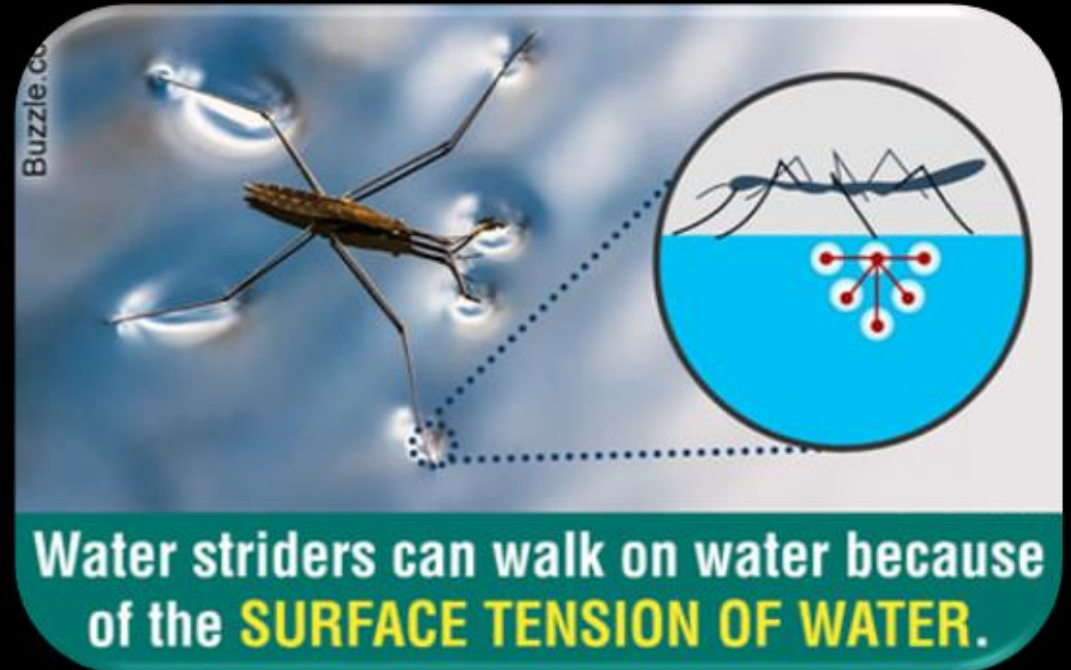
- Cohesion is the tendency of molecules within a substance to 'stick together'.
- Water has a high Cohesion because of Hydrogen bonding.
- This is important as transport of water in the xylem in plants relies on water being pulled up.
- Cohesion also gives the water a high surface tension, allowing small organisms, such as Pond Skaters, to walk along it.



Properties of water:

HIGH SURFACE TENSION

- Water has high surface tension because of the attraction between the water molecules.
- The attraction of hydrogen bonds between water molecules is greater than the attraction between the water molecules and other molecules in air.



High surface tension



▲ **fig 1** Without surface tension, a raft spider like this could not move across the water and hunt.

High surface tension means that the water surface acts as a “skin”; this is very important in water being drawn up the phloem of a plant and for life at the surface of ponds/lakes.

SUMMARY: Solvent Properties of Water

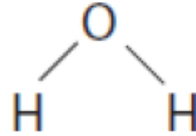
- Water is a good transport medium as it is a polar solvent.
- Water; because of its dipolar nature, surrounds ionic/polar molecules and form H- bonds, dissolving them.
- It is a liquid hence is able to flow due to the cohesive and adhesive forces.
- Water allows transport of polar substances like glucose and amino acids in mass transport in blood of animals.
- Water allows transport of inorganic ions from roots to leaves in plants via xylem vessels.
- Water acts as an aqueous medium to allow chemical reactions to takes place in cell cytoplasm.
- Water act as a reactant in important metabolic reactions in living organisms.
Eg: Hydrolysis & Photosynthesis

Command Words

1. Water is a polar molecule. The diagram shows a molecule of water.

(a) Complete the diagram to show the dipole nature of this water molecule.

(2)



COMPLETE/RECORD

Requires the completion of a table/diagram/equation.

Command Words

(b) **Name** the type of reaction in which a molecule of water is involved in the breaking of a bond in another molecule. (1)

.....

GIVE/STATE/NAME

All of these command words are really synonyms. They generally all require recall of one or more pieces of information.

Command Words

(c) **Explain** how the properties of water make it an ideal transport medium.

(3)

EXPLAIN

An explanation requires a justification/exemplification of a point.
The answer must contain some element of reasoning/justification;
this can include mathematical explanations.

Checkpoint Questions

1. Water is described as a dipolar molecule because it has a _____ (1)

- ☐ A positively charged hydrogen end and a negatively charged oxygen end
- ☐ B positively charged hydrogen end and a positively charged oxygen end
- ☐ C negatively charged hydrogen end and a negatively charged oxygen end
- ☐ D negatively charged hydrogen end and a positively charged oxygen end

ANSWER: A

Checkpoint Questions

2. Fill in the blanks

Water has the chemical formula _____. Water molecules are described as _____ because they have a slight positive charge at one end of the molecule and a slight negative charge at the other end. As a result, individual molecules form _____ bonds with each other. Water is an important _____ in living organisms because most biochemical reactions take place in aqueous solution. Water also has a high _____ which means that its temperature remains relatively stable despite large changes in the temperature of the surrounding environment.

Checkpoint Questions

2.

ANSWER:

- H_2O [1]
- Bipolar / Dipolar / polar [1]
- Weak hydrogen / H – bonds [1]
- Solvent [1]
- Specific heat capacity [1]

Checkpoint Questions

3. What is the name between the bond between two water molecules? (1)

ANSWER:

Hydrogen bond

4. State the property of water that enable these bonds to be formed. (1)

ANSWER:

Dipole nature

Checkpoint Questions

6. Explain why water is an effective molecule for transporting other molecules around living organisms. (3)

ANSWER:

- water is a solvent
- water is slightly charged / dipole / polar
- other polar molecules or ions can dissolve in water
- water is a fluid with cohesive / adhesive forces

EXAM HINT

Make sure you understand and can explain the difference between ionic substances, charged particles and polar molecules.

EXAM HINT

All these properties are a result of dipoles and hydrogen bonding between water molecules. Make sure that you can explain the link between the property and the hydrogen bonding.

EXAM HINT

In exam questions, the command word *discuss* suggests that you may need to consider possible negative effects as well as the more obvious benefits to living organisms. You should identify the issue that is being assessed within the question. Explore all aspects of the issue. Investigate the issue by reasoning or argument.

LEARNING TIP

Remember that *co* means two similar things together, as in cohabit, and *ad* means two different things together.

SUBJECT VOCABULARY

anion a negative ion

cation a positive ion

ionic bonds bonds formed when atoms give or receive electrons; they result in charged particles called ions

covalent bonds bonds formed when atoms share electrons; covalent molecules may be polar if the electrons are not shared equally

dipole the separation of charge in a molecule when the electrons in covalent bonds are not evenly shared

polar molecule a molecule containing a dipole

dissociation splitting of a molecule into smaller molecules, atoms, or ions, especially by a reversible process

hydrogen bonds weak electrostatic intermolecular bonds formed between polar molecules containing at least one hydrogen atom